



BIO-CHEM TEST SERIES

(Academic Session : 2023 - 2024)

NEET(UG)
Bio-Chemistry
15-02-2024

**PRE-MEDICAL : MEA+MEB+MEC+JMEA+MLA+MLB+MLC+MLD+
JMLA +JMLD+ MAP + MAQ + MAS + MPA + SMAR**

This Booklet contains 20 pages.

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ENTHUSIAST, LEADER & ACHIEVER COURSE (ALL PHASE)

BIO-CHEMISTRY TEST

DATE : 15 - 02 - 2024

SYLLABUS – 01

CHEMISTRY : TG: @NEETxNOAH Chemical Equilibrium & Ionic Equilibrium

BIOLOGY : The Living World, Biological Classification, Plant Kingdom, Animal Kingdom

1. What is wrong about equilibrium state?

- (1) $\Delta G = 0$
- (2) The reaction ceases at equilibrium
- (3) Equilibrium constant is independent of initial concentrations of reactants
- (4) Catalyst has no effect on equilibrium state.

2. The equilibrium constant for the reaction $\text{NH}_4\text{NO}_2(\text{s}) \rightleftharpoons \text{N}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$, is given by

(1) $\frac{[\text{NH}_4\text{NO}_2]}{[\text{N}_2][\text{H}_2\text{O}]^2}$

(2) $[\text{N}_2][\text{H}_2\text{O}]^2$

(3) $\frac{[\text{N}_2][\text{H}_2\text{O}]}{[\text{NH}_4\text{NO}_2]}$

(4) $\frac{[\text{N}_2][2\text{H}_2\text{O}]^2}{[\text{NH}_4\text{NO}_2]^2}$

3. The equilibrium constant for a reaction $\text{A} + 2\text{B} \rightleftharpoons 2\text{C}$ is 40 then equilibrium constant

for reaction $\text{C} \rightleftharpoons \text{B} + \frac{1}{2}\text{A}$ will be :

(1) $1/40$ (2) $(1/40)^{1/2}$

(3) $(1/40)^2$ (4) 40

4. The equilibrium constant (K) for the reaction: $2\text{HI}(\text{g}) \rightleftharpoons \text{H}_2(\text{g}) + \text{I}_2(\text{g})$ at room temperature is 2.85 and that of 698 K is 1.4×10^{-2} . This implies that the forward reaction is

- (1) Exothermic
- (2) Endothermic
- (3) Exergonic
- (4) Unpredictable

5. For the reaction : $2\text{A}(\text{g}) + \text{B}(\text{g}) \rightleftharpoons 3\text{C}(\text{g}) + 4\text{D}(\text{g})$
Two moles each of A and B were taken into a 1L flask. The following must always be true when the system attained equilibrium

- (1) $[\text{A}] = [\text{B}]$
- (2) $[\text{A}] < [\text{B}]$
- (3) $[\text{B}] = [\text{C}]$
- (4) $[\text{A}] + [\text{B}] < [\text{C}] + [\text{D}]$.

6. In which of the following reaction K_p and K_c are equal

- (1) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$
- (2) $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$
- (3) $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$
- (4) $2\text{NO}(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$

7. For a hypothetical reaction : $\text{AB}_2(\text{g}) + \frac{1}{2}\text{B}_2(\text{g}) \rightleftharpoons \text{AB}_3(\text{g})$; $\Delta H = -x \text{ kJ}$. More AB_3 could be produced at equilibrium by :

- (1) using a catalyst
- (2) removing some of B_2
- (3) increasing the temperature
- (4) increasing the pressure.

8. The rate constants of forward and backward reactions are 8.5×10^{-5} and 2.38×10^{-4} respectively. The equilibrium constant is

- (1) 0.35
- (2) 0.42
- (3) 12.92
- (4) 0.292.

9. The vapour density of N_2O_4 at a certain temperature is 30. What is the percentage dissociation of N_2O_4 at this temperature?

- (1) 53.3%
- (2) 106.6%
- (3) 26.7%
- (4) None.

10. For the following equilibrium
 $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$
 K_p is found to be equal to K_c . This is attained when:
 (1) $T = 1 \text{ K}$
 (2) $T = 12.18 \text{ K}$
 (3) $T = 27.3 \text{ K}$
 (4) $T = 273 \text{ K}$.
11. 1 mole each of $\text{CO}(\text{g})$, $\text{H}_2\text{O}(\text{g})$, $\text{H}_2(\text{g})$ and $\text{CO}_2(\text{g})$ are placed in one litre flask at 25°C . When following equilibrium is set-up
 $\text{CO}(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{CO}_2(\text{g}) + \text{H}_2(\text{g}); K_p = 9$.
 Hence CO_2 present at equilibrium is:
 (1) 0.5 mol
 (2) 1.5 mol
 (3) 0.25 mol
 (4) 3.0 mol.
12. One mole of $\text{N}_2\text{O}_4(\text{g})$ is kept in a closed container under one atmosphere at 300 K. When it is heated to 600 K then 20% of $\text{N}_2\text{O}_4(\text{g})$ is converted to $\text{NO}_2(\text{g})$ according to given reaction
 $\text{N}_2\text{O}_4 \rightleftharpoons 2\text{NO}_2(\text{g})$,
 then resultant pressure at equilibrium is:
 (1) 1.2 atm (2) 2.4 atm
 (3) 2.0 atm (4) 1.0 atm.
13. For reaction : $\text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g}) \rightleftharpoons \text{PCl}_5(\text{g})$
 the value of K_c at 250°C is 26 then value of K_p at this temperature will be
 (1) 0.6 (2) 5.7
 (3) 8.3 (4) 4.6
14. If K_p for a reaction: $\text{A}(\text{g}) + 2\text{B}(\text{g}) \rightleftharpoons 3\text{C}(\text{g}) + \text{D}(\text{g})$ is 0.05 atm at 1000 K then its K_c in terms of R will be
 (1) 20000 R (2) 0.02 R
 (3) $5 \times 10^{-5} \text{ R}$ (4) $\frac{5 \times 10^{-5}}{\text{R}}$
15. In which of the following cases, the reaction goes near to completion?
 (1) $K = 10^3$ (2) $K = 10^{-2}$
 (3) $K = 10$ (4) $K = 10^0$.
16. For a reaction: $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$, the value of equilibrium constant is 50 at 271 K. If 0.5 moles each of H_2 and I_2 is added to the reaction system, then value of equilibrium constant will be
 (1) 0.02 (2) 0.2
 (3) 50 (4) 25.
17. 1.1 mol of A is mixed with 2.2 mol of B and the mixture is kept in one litre flask till the equilibrium is reached. If 0.2 mol of C is formed at equilibrium then value of equilibrium constant for the given reaction $\text{A} + 2\text{B} \rightarrow 2\text{C} + \text{D}$ is
 (1) 0.002 (2) 0.004
 (3) 0.001 (4) 0.003.
18. In which of the following reaction $K_p > K_c$?
 (1) $\text{N}_2(\text{g}) + 2\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$
 (2) $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$
 (3) $\text{PCl}_3(\text{g}) + \text{Cl}_2 \rightleftharpoons \text{PCl}_5(\text{g})$
 (4) $2\text{SO}_3(\text{g}) \rightleftharpoons 2\text{SO}_2(\text{g}) + \text{O}_2(\text{g})$.
19. For the reaction : $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$, the forward reaction at constant temperature is favoured by :
 (1) introduction of an inert gas at constant volume
 (2) introduction of $\text{PCl}_3(\text{g})$ at constant volume
 (3) introduction of $\text{PCl}_5(\text{g})$ at constant volume
 (4) introduction of Cl_2 at constant volume.
20. In the case of gaseous homogeneous reaction, the active mass of the reactant is obtained by the expression
 (1) $\frac{PV}{RT}$ (2) $\frac{P}{RT}$
 (3) $\frac{RT}{P}$ (4) $\frac{n}{V}RT$

21. The chemical equilibrium of a reversible reaction is not influenced by
- pressure
 - catalyst
 - concentration of the reactants
 - temperature.
22. $\text{CaCO}_3 \rightleftharpoons \text{CaO} + \text{CO}_2$ the given reaction goes to completion in a lime kiln because
- CaO does not react with CO_2 to give CaCO_3
 - backward reaction is very slow
 - CO_2 formed escapes out
 - None of these
23. The K_{sp} of Ag_2CrO_4 is 1.1×10^{-12} at 298 K then solubility (in mol/L) of Ag_2CrO_4 in 0.1 M AgNO_3 solution will be
- 1.1×10^{-11}
 - 1.1×10^{-10}
 - 1.1×10^{-12}
 - 1.1×10^9
24. The Bronsted acid which gives the weakest conjugate base is
- HF
 - H_2S
 - HCl
 - H_2O .
25. Which of the following is a Lewis acid?
- H_2O
 - SnCl_4
 - $\text{C}_2\text{H}_5\text{OH}$
 - Cl^- .
26. NH_4OH is a weak base but it becomes still weaker in the aqueous solution of
- 0.1 M HCl
 - 0.1 M NH_4Cl
 - 0.1 M H_2SO_4
 - 0.1 M CH_3COOH .
27. The addition of NaCl to AgCl decreases the solubility of AgCl because
- Solubility product decreases
 - Solubility product remains constant
 - Solution becomes unsaturated
 - Solution becomes supersaturated.
28. Which pair will show common ion effect?
- $\text{BaCl}_2 + \text{Ba}(\text{NO}_3)_2$
 - $\text{NaCl} + \text{HCl}$
 - $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$
 - $\text{AgCN} + \text{KCN}$.
29. Which of the following has maximum pH?
- 1N CH_3COOH
 - 1N HCl
 - 1N H_2SO_4
 - 1N HNO_3 .
30. The value of pK_w of water
- increases with increase in temperature
 - decreases with increase in temperature
 - does not change with variation in temperature
 - increases till 50°C and there after decreases.
31. When 2 moles of HCl is added to 1 litre of an acidic buffer, its pH changes from 3.4 to 2.9. The buffer capacity of the buffer solution is
- 2
 - 0
 - 4
 - 8
32. The pH of a dilute solution of acetic acid was found to be 4.3 then addition of a small crystal of sodium acetate will cause pH to
- become less than 4.3
 - become more than 4.3
 - remain equal to 4.3
 - unpredictable.

33. Which of the following is not a buffer solution?
 (1) $\text{CH}_3\text{COOH} + \text{CH}_3\text{COONa}$
 (2) $\text{HCl} + \text{NaCl}$
 (3) $\text{HCOOH} + \text{HCOONa}$
 (4) $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$
34. Which indicator should be used in titrating KOH against oxalic acid?
 (1) Litmus (2) Methyl orange
 (3) Methyl red (4) Phenolphthalein.
35. Three sparingly soluble salts M_2X , MX and MX_3 have same value of solubility product. Their solubilities follow the order
 (1) $\text{MX}_3 > \text{MX} > \text{M}_2\text{X}$
 (2) $\text{MX} > \text{MX}_3 > \text{M}_2\text{X}$
 (3) $\text{MX} > \text{M}_2\text{X} > \text{MX}_3$
 (4) $\text{MX}_3 > \text{M}_2\text{X} > \text{MX}$.
36. Solubility of a salt A_2B_3 is $1 \times 10^{-4} \text{ mol L}^{-1}$ then its solubility product is
 (1) 1.08×10^{20} (2) 1.08×10^{18}
 (3) 2.6×10^{-18} (4) 1.08×10^{-18} .
37. If K_a of $\text{HCN} = 4 \times 10^{-10}$ then the pH of 2.5×10^{-1} molar HCN(aq) is
 (1) 2×10^{-2} (2) 4.7
 (3) 2.0 (4) 5.0
38. The pH of 10^{-10} M NaOH solution at 25°C is approximately TG: @NEETxNOAH
 (1) 7
 (2) 10
 (3) between 10 and 11
 (4) between 6 and 7.
39. The dissociation constant of a weak acid HA is 1×10^{-4} then equilibrium constant for its reaction with strong base BOH is
 (1) 1×10^{-4} (2) 1×10^{-10}
 (3) ∞ (4) 1×10^{10}
40. K_a for HCN is 5×10^{-10} at 25°C then how much volume of 5 M KCN solution is required to be added to 10 mL of 2 M HCN in order to maintain $\text{pH} = 9$.
 (1) 2 mL (2) 5 mL (3) 3 mL (4) 4 mL .
41. pK_b of NH_3 is 4.74. Then what will be the pH when 100 mL of 0.01 M NH_3 solution is 50% neutralised by 0.01 M HCl
 (1) 4.74 (2) 2.37
 (3) 9.26 (4) 9.48
42. The solubility of PbCl_2 is
 (1) $\sqrt{K_{sp}}$
 (2) $(K_{sp})^{1/3}$
 (3) $(K_{sp}/4)^{1/3}$
 (4) $(8K_{sp})^{1/2}$
43. If a neutral solution has $\text{pK}_w = 13.36$ at 50°C , then pH of the solution is
 (1) 6.68 (2) 7
 (3) 7.68 (4) None of these.
44. The pH value of $\text{M}/1000$ solution of KOH in water is
 (1) 3 (2) 6
 (3) 11 (4) 9
45. **Assertion (A)** : For equilibrium $\text{X}_{(g)} \rightleftharpoons \text{Y}_{(g)}$ total no. of moles at new equilibrium is less than old equilibrium if some amount of $\text{X}_{(g)}$ is removed.
Reason (R) : The no. of moles of the substance removed is partially compensated as the system reaches to new equilibrium.
 (1) Both A and R are true and R is the correct explanation of A.
 (2) Both A and R are true and R is not a correct explanation of A.
 (3) A is true but R is false
 (4) A is false but R is true.

46. Polysulphate ester is present in the cell wall of A which also show B. What are A and B respectively:
- A = Green algae, B = A granal chloroplast
 - A = Red algae, B = Isogamy
 - A = Brown algae, B = Heterotrichous thallus
 - A = Red Algae, B = Advanced type of sexual reproduction
47. Which of the following option correctly identifies the plant shown in figure and group it belongs to?



- Marchantia - a liverwort
 - Sphagnum - a moss
 - Sphgnum - a Liverwort
 - Funaria - a moss
48. Red algae differ from the brown algae in having:
- Chlorophyll 'a'
 - Aquatic nature
 - Cellulosic cell wall
 - Floridean starch
49. Pyrenoids are
- Core of starch surrounded by protein
 - Core of protein surrounded by fats
 - Proteinaceous centre and surrounded by starch
 - Core of nucleic acid surrounded by protein
50. Classification of algae is mainly based on:-
- Type of pigments
 - Stored food
 - Type of reproduction
 - Cell wall

51. Which of the following statement is correct?
- Brown algae possess chlorophyll a, b carotenoids, and xanthophyll.
 - Majority of red algae are found in fresh water
 - Volvox, Fucus and Ulothrix are green algae
 - Sexual Reproduction may be isogamous, anisogamous or oogamous in brown algae.
52. Antherozoid of bryophytes are usually:-
- Non-flagellated coma shaped
 - Biflagellated, coma shaped
 - Multiflagellated, spiral
 - Monoflagellated, spiral
53. Select the incorrect statement from the followings:-
- The plant Body of liverworts is not thalloid but made up of rhizoids, stem and leaves.
 - Mosses have upright, slender axis bearing spirally arranged leaves like structure
 - Spores germinate to form gametophyte in liverwort
 - The zygote produces a sporophyte in embryophyta
54. Which one of following statement regarding bryophytes is correct?
- The plant body is less differentiated than that of algae
 - They have true roots and leaves but not stem
 - Antherozoids are mainly multiciliated
 - Zyotes do not undergo reduction division immediately

TG: @NEETxNOAH

55. Read the following statements:-

- (i) The predominant stage of life cycle is the gametophyte
- (ii) The sex organs antheridia and archegonia are produced at the apices of the leafy shoot
- (iii) Vegetative reproduction is by fragmentation and budding in the secondary protonema

Which of the following is correctly matched with all the above given statement (i-iii)?

Option:-

- (1) Riccia
- (2) Selaginella
- (3) Chara
- (4) Funaria

56. Bryophytes resemble algae in:-

- (1) Differentiation of the plant body into root, stem and heterotrophic mode of nutrition
- (2) Gametophytic like plant body, lack of vascular tissue, absence of roots and autotrophic mode of nutrition
- (3) Thallus like plant body, presence of roots and heterotrophic mode of nutrition
- (4) Filamentous body, presence of vascular tissue and autotrophic mode of nutrition.

57. What is common among Porphyra, Agaricus, Chlorella and Spirulina?

- (1) They all are photosynthetic
- (2) They all have eukaryotic cell organization
- (3) They all are single celled
- (4) They all either used as food or food supplements

58. Match the following Column-I and Column-II :-

	Column-I		Column-II
A	Ulothrix	i	Isogamy
B	Fucus	ii	Diplontic
C	Polysiphonia	iii	Oogamy

- (1) A-i, B-ii, C-iii
- (2) A-i, B-iii, C-ii
- (3) A-ii, B-iii, C-i
- (4) A-iii, C-ii, C-i

59. Which one of the following statements is true?

- (1) Agar-agar is obtained from Gelidium and Gracilaria
- (2) Laminaria and Sargassum are used as food
- (3) Algae increase the level of dissolved oxygen in the immediate environment
- (4) All of the above

60. Read the following statements carefully and identify two correct statements:-

- (A) In brown algae fusion of gametes occurs either in water or inside oogonium
- (B) Pigments of blue green algae are similar to brown algae
- (C) Chlorophyll a & d are present in red algae
- (D) Haplontic life cycle is observed in algae & bryophyta

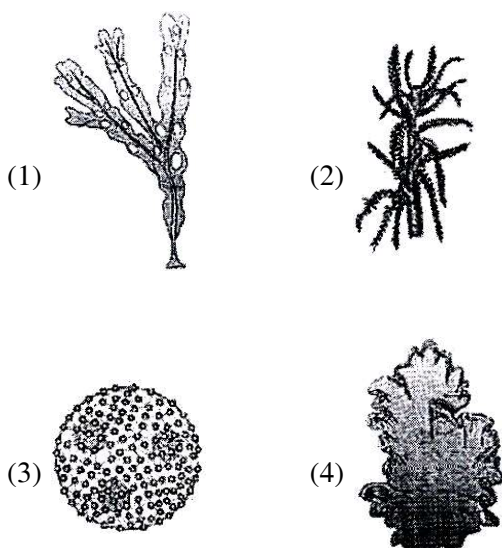
Option:-

- (1) B and C
- (2) A and B
- (3) A and C
- (4) A and D

61. Which of the following statement is incorrect for Algae?

- (1) Two, unequal lateral flagella are found in brown algae
- (2) Chlorophyll-a is present in green, brown and red algae
- (3) The red algae usually reproduce vegetatively by fragmentation
- (4) Zoospores are found in red algae

62. Mosses are ecologically important as:-
 (1) They are source of absorbant cotton
 (2) They reduces soil erosion
 (3) Mosses are first organism to colonise water lake
 (4) All of the above
63. Which of the following statement is correct?
 (1) In mosses asexual reproduction takes place by green, unicellular buds called gammae
 (2) The sporophyte of moss is differentiated in foot, seta and capsule
 (3) Protonema is a green, creeping, branched and filamentous stage in red algae, which develops directly from gametes.
 (4) Moss have an elaborate mechanism of seed dispresal.
64. In the following diagrams which one provides algin that have long been used in ice cream?



65. Which of the following algae show diplontic life cycle?
 (1) Ulothrix
 (2) Spirogyra
 (3) Ectocarpus
 (4) Fucus

66. Find the correct about algae
 (1) They have vascular tissues
 (2) They produce antheridia as female sex organ
 (3) They produce archegonia as female sex organ
 (4) They produce oogonia as female sex organ/structure
67. Which of the following pair is an example of heterosporous pteridophyte ?
 (1) *Psilotum* and *Adiantum*
 (2) *Salvinia* and *Selaginella*
 (3) *Lycopodium* and *Dryopteris*
 (4) *Pteris* and *Adiantum*
68. In pteridophyte gametophyte (prothallus) prefers to grow in moist shady places because :
 (1) In these conditions, it survive very well
 (2) It represents the ancestry of bryophytes
 (3) It need water for fertilization
 (4) In pteridophytes roots are short
69. Consider the following four statements (a-d) and select the option which include the all correct ones only
 (a) Unlike bryophytes and pteridophytes, in gymnosperms the male and female gametophyte do not have an independent free living existence.
 (b) In pteridophytes, the female gametophyte is retained on the parent sporophyte for variable period.
 (c) Zygote is the first cell of sporophyte generation
 (d) Bryophytes do not play any role in plant succession occuring on bare rock or soil or Terrestrial habitate.
- (1) b, c and d
 (2) a, c and d
 (3) a, b and d
 (4) a, b and c

70. Select the true statement from the following :-

- (1) Kingdom-Animalia is characterised by heterotrophic eukaryotic organisms that are multicellular and their cells lack cell walls
- (2) Mode of nutrition in members of Kingdom Animalia is holozoic and stored food is glycogen
- (3) A few members of Kingdom-Plantae are partially heterotrophic or parasites
- (4) All of these

71. Both *Chlamydomonas* and *Chlorella* are autotrophic and both have cell wall whereas both *Paramecium* and *Amoeba* are heterotrophic and both lack cell wall, then why R.H. Whittaker place all of them under the same kingdom ?

- (1) Because all of them are single celled organisms.
- (2) Because all of them have nuclear membrane and other membrane bound cell organelles
- (3) Because all of them have cell membrane
- (4) Both (1) & (2)

72. Choose the incorrect statement :-

- (1) The vast majority of bacteria are heterotrophs
- (2) Chemosynthetic bacteria are example of Autotrophs
- (3) Nucleus is present in all nitrifying bacteria
- (4) Bacterial structure is very simple and they are very complex in behaviour

73. Choose the incorrect statement of following :-

- (1) Ribosomes are the site of protein synthesis.
- (2) Pili do not play a role in motility.
- (3) Inclusion bodies are bounded by membrane.
- (4) Cell wall prevent the bacterium from bursting and collapsing.

74. Consider the following four statements (A-D) and select the option which includes all the correct ones only.

- (A) Archaeobacteria differ from other bacteria in having a different cell wall structure
- (B) Bacteria as a group show the most extensive metabolic diversity
- (C) Eubacteria have chlorophyll 'c' similar to green plants.
- (D) Mycoplasma can survive without oxygen.

Option :-

- (1) Statements B, C and D
- (2) Statements A, C
- (3) Statements B, D
- (4) Statements A, B and D

75. How many organism are included in Monera in given list ?

- | | |
|-------------------|---------------------|
| (A) Slimemoulds | (B) Dinoflagellates |
| (C) Protozoans | (D) Pseudomonas |
| (E) Archebacteria | (F) Basidiomycetes |

- | | |
|-------|-------|
| (1) 2 | (2) 4 |
| (3) 2 | (4) 5 |

76. Which of the following is not related with protista ?

- (1) Eukaryote
- (2) Membrane bound organelles
- (3) Both sexual and asexual reproduction
- (4) Kingdom protista has well define boundary regarding classification

77. Which option is not correct about members of kingdom protista ?

- (1) They have cilia or flagella
- (2) They secrete poison in water
- (3) They show nitrogen fixation
- (4) They have well defined nucleus and other membrane-bounded cell organelles

78. Which of the following pigment is present in Euglenoid but lack in dinoflagellates and diatoms:-

- (1) Chlorophyll 'a'
- (2) Chlorophyll 'c'
- (3) Chlorophyll 'b'
- (4) Caroteens

79. Select out the incorrect match :-

- (1) Euglenoid-fresh stagnant water
- (2) Diatoms-Marine water
- (3) Dinoflagellates-Damp soil
- (4) Slime mould-treelogs and plant waste

80. Mycelium of basidiomycetes is :

- (1) Branched and septate
- (2) Unbranched and aseptate
- (3) Branched and coenocytic
- (4) Unbranched and coenocytic

81. Identify the correct match from the columns I, II and III :-

Column I		Column II		Column III	
(1)	Albugo	(a)	Ascomycetes	(i)	Obligate parasite
(2)	Agaricus	(b)	Deuteromycetes	(ii)	Decomposer of litter
(3)	Aspergillus	(c)	Phycomycetes	(iii)	Branched and septate mycelium
(4)	Alternaria	(d)	Basidiomycetes	(iv)	edible fungus

- (1) 1-c-i, 2-a-iv, 3-a-iii, 4-b-ii
- (2) 1-c-i, 2-d-iv, 3-d-iii, 4-a-ii
- (3) 1-c-i, 2-a-iii, 3-d-iv, 4-b-ii
- (4) 1-c-i, 2-d-iv, 3-a-iii, 4-b-ii

82. Choose the incorrect statement about phycomycetes

- (1) Members are found in aquatic habitats
- (2) Spores are endogenously produced in sporangium
- (3) Asexual spore are conidia
- (4) They show all type of syngamy

83. Which statement is not associated with fungi :-

- (1) Cell wall is composed of Chitin and polysaccharides
- (2) Show symbiotic association with Algae
- (3) They are multicellular eukaryotic autotrophic
- (4) They causes smut, late blight & rot diseases

84. Choose the incorrect statement regarding basidiomycetes :-

- (1) Asexual reproduction with the help of fragmentation.
- (2) Rust, smut and mushroom are related with this group
- (3) Karyogamy and meiosis take place in the basidium.
- (4) Commonly known forms are *Agaricus*, *Aspergillus* and *Alternaria*.

85. Which of the following statement is incorrect ?

- (1) Fungi prefer to grow in cold and humid places
- (2) Karyogamy in fungi occurs after plasmogamy
- (3) Yeast are used to make bread and beer.
- (4) Mycelium consist of long, slender thread like structures called fungal hyphae.

86. Which one of the following is mismatched :-

- (1) *Penicillium* - Source of antibiotics
- (2) *Albugo* - Parasitic fungi on legume
- (3) *Neurospora* - Used in biochemical and genetic work
- (4) *Agaricus* - Edible fungus

87. **Statement-1** : Lichens are parasitic association.

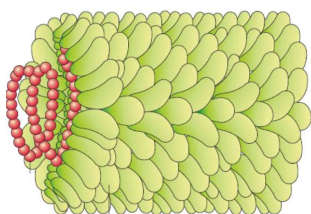
Statement-2 : In Lichens, algae prepare food for fungi and fungi provide shelter and absorb mineral nutrients and water for its partner.

- (1) Statement-1 is true, statement -2 is true; Statement -2 is not the correct explanation of statement -1
- (2) Statement-1 is true, statement -2 is true; Statement -2 is the correct explanation of statement -1
- (3) Statement-1 is true, statement -2 is false
- (4) Statement-1 is false, statement -2 is true

88. Infectious and free RNA particle having low molecular weight were discovered by :-

- (1) Diener
- (2) Beijerinck
- (3) Stanley
- (4) Pasture

89. How many statement are/is correct regarding given diagram ?



- (a) Virus has genetic material DNA.
 - (b) Virus has genetic material RNA.
 - (c) Virus has genetic material Protein.
 - (d) It causes infection in animal.
 - (e) They are living in a dicot plant.
- (1) 2 (2) 3 (3) 4 (4) All

90. Find out the match - that is not correct for given organisms -

- (1) *Mycoplasma* - can survive without oxygen
- (2) *Dinoflagellates* - soap box like cell walls
- (3) *Euglenoids* - Majority of them are fresh water organisms
- (4) *Diatoms* - float passively on water surface

91. Which of the following is gregarious pest:

- (1) *Locusta*
- (2) *Laccifer*
- (3) *Aedes*
- (4) *Limulus*

92. Which of the following structure is found in phylum Mollusca:

- (1) Radula
- (2) Malpighian Tubules
- (3) Nephridia
- (4) All of the above

93. Space between mantle and visceral hump contains feather like structure called gills, function of these gill is/are:

- (1) Respiration
- (2) Digestion
- (3) Reproduction
- (4) (1) and (2) both

94. Proboscis gland is present in:

- (1) *Doliolum*
- (2) *Salpa*
- (3) *Branchiostoma*
- (4) *Balanoglossus*

95. Incorrect feature of *Lamprey* is:

- (1) 6-15 pair's of gill slits present
- (2) Bony vertebral column
- (3) Circulation is close
- (4) Sucking and circular mouth

96. Repere word is related to

- (1) Reptilia
- (2) Amphibia
- (3) Aves
- (4) Mammalia

97. Bird is:

- (1) Poikilothermic
- (2) Homoiothermous
- (3) Fossorial
- (4) Parasite

98. Oil gland is present in:

- (1) Cockroach
- (2) Coelenterata
- (3) Aves
- (4) Sponge

99. Water vascular system is the characteristic feature of:

- (1) Porifera
- (2) Echinodermata
- (3) Urochordata
- (4) Coelenterata

100. Second largest phylum of animal kingdom is:

- (1) Arthropoda
- (2) Mollusca
- (3) Echinodermata
- (4) Chordata

101. Type of symmetry present in Echinodermata (adult).

- (1) Asymmetry
- (2) Radial
- (3) Bilateral
- (4) None of these

102. What is common between earthworm and cockroach:

- (1) Nephridia
- (2) Ventral nerve cord
- (3) Closed circulation
- (4) Malpighian tubules

103. Pearl is obtained from:

- (1) Asterias
- (2) *Pinctada*
- (3) Rohu
- (4) Octopus

104. Echinodermata is a group of animal which are:

- (1) Coelomate, spiny and calcareous ossicles
- (2) Acoelomate, spiny and marine calcareous ossicles
- (3) Acoelomate, horny and marine calcareous ossicles
- (4) Jointed legs, coelomate and marine calcareous ossicles

105. A flying fish is:

- (1) Rohu
- (2) Magur
- (3) Exocoetus
- (4) *Trygon*

106. A marine cartilagenous fish that can produce electric current is:

- (1) *Pristis*
- (2) *Torpedo*
- (3) *Trygon*
- (4) *Scoliodon*

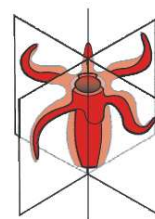
107. Which of the following is true fish:

- (1) Jelly fish
- (2) Star fish
- (3) Dog fish
- (4) Devil fish

108. Scales are absent in:

- (1) Lamprey
- (2) Cartilagenous fish
- (3) Bony fish
- (4) Reptilia

109. Following diagram represent which type of symmetry



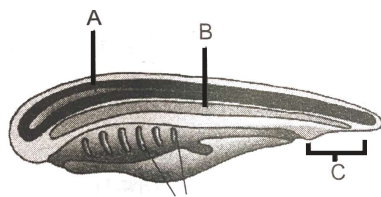
- (1) Asymmetry
- (2) Radial symmetry
- (3) Bilateral symmetry
- (4) None of these

110. The given figure represents :



- (1) Tapeworm
- (2) Round worm
- (3) Flat worm
- (4) Both (1) and (3)

111. Choose the correct remark A,B and C in given diagram



	(B)	(A)	(C)
(1)	Notochord	Nerve cord	Post anal part
(2)	Nerve cord	Notochord	Post anal part
(3)	Notochord	Vertebral column	Gill slits
(4)	Nerve cord	Vertebral column	Gill slits

112. Consider the following statements.

- (A) Mammals are exclusively marine
- (B) In urochordates, notochord extends from head to tail region
- (C) In cephalochordates, notochord is present only in larval tail

Which of the above statement is/are incorrect?

- (1) A alone
- (2) Only A and B
- (3) Only B and C
- (4) All of these

113. Match the items in column-I with those in column-II and choose the correct option.

Column-I

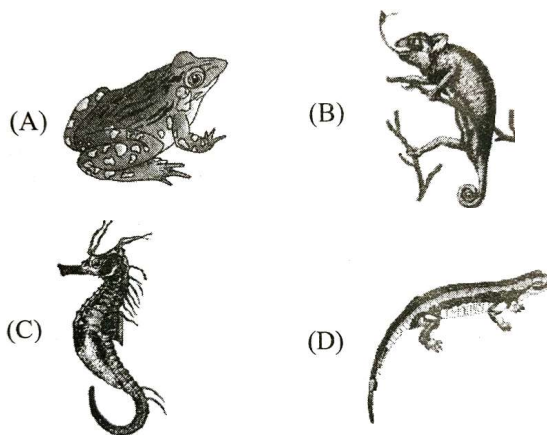
Column-II

- (A) *Chelone*
- (B) *Vipera*
- (C) *Euspongia*
- (D) *Bangarus*

- (1) Wall lizard
- (2) Viper
- (3) Krait
- (4) Scypha
- (5) Turtle
- (6) Bath sponge

- (1) A - 3, B - 2, C - 1, D - 6
- (2) A - 5, B - 2, C - 4, D - 1
- (3) A - 5, B - 4, C - 1, D - 6
- (4) A - 5, B - 2, C - 6, D - 3

114. Which of the following are amphibians ?



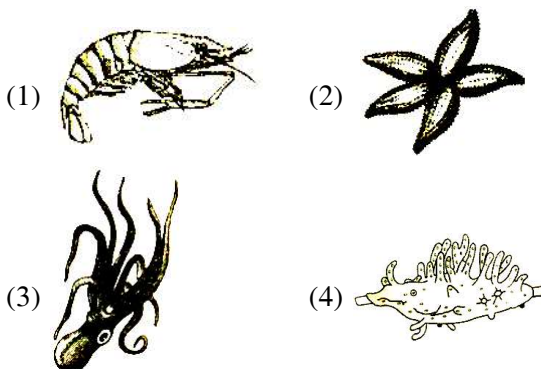
- (1) A and C
- (2) A and D
- (3) C and D
- (4) B and C

115. Identify the figure with its correct name and phylum.



- (1) Balanoglossus - Urochordata
- (2) Ascidia - Urochordata
- (3) Balanoglossus - Hemichordata
- (4) Hirudinaria - Annelida

116. Which of the following animal's body is covered by calcareous shell and unsegmented with a distinct head, muscular foot, and visceral hump?



117. Flame cells present in platyhelminthes are specialized in-

- (1) Respiration and absorption.
- (2) Osmoregulation and circulation.
- (3) Respiration and excretion.
- (4) Osmoregulation and excretion.

118. Match the organisms given in column-I with their common name given in column-II and choose the correct option.

Column-I**(Organisms)**

A. Pennatula

B. Gorgonia

C. Adamsia

D. Cucumaria

Column-II**(Common name)**

I. Sea-fan

II. Sea-pen

III. Sea-Anemone

IV. Sea-cucumber

(1) A-II; B-III; C-I; D-IV

(2) A-II; B-IV; C-I; D-III

(3) A-II; B-I; C-III; D-IV

(4) A-II; B-I; C-IV; D-III

119. Which of the following possesses poison sting and belongs to class chondrichthyes?

- (1) *Labeo*
- (2) *Myxine*
- (3) *Clarias*
- (4) *Trygon*

120. **Statement I :** Hemichordata was earlier considered as a sub-phylum under phylum Chordata.

Statement II : Balanoglossus and Saccoglossus are urochordates.

- (1) Both statements are correct
- (2) Statement I is correct & II incorrect
- (3) Statement I is incorrect & II correct
- (4) Both statements are incorrect

121. **Statement I :** All vertebrates are chordates and all chordates are vertebrates.

Statement II : The notochord is replaced by a cartilaginous or bony vertebral column in the vertebrates.

- (1) Both statements are correct
- (2) Statement I is correct & II incorrect
- (3) Statement I is incorrect & II correct
- (4) Both statements are incorrect

122. **Statement I :** Amphibia, Reptilia, Aves and Mammals included in super class tetrapoda.

Statement II : They have two pairs of limbs, adapted for walking, running, climbing, burrowing, swimming or flying.

- (1) Both statements are correct
- (2) Statement I is correct & II incorrect
- (3) Statement I is incorrect & II correct
- (4) Both statements are incorrect

123. **Assertion :** All members of animalia are multicellular.

Reason : All of them exhibit the same pattern of organisation of cells.

- (1) Assertion and reason both are wrong.
- (2) Assertion and reason both are true but reason is not correct explanation of assertion.
- (3) Assertion is true but reason is wrong.
- (4) Assertion and reason both are true and the reason is correct explanation of assertion.

124. Assertion : Sponges exhibit cellular level of organisation.

Reason : In sponge cells are arranged as loose cell aggregates.

- (1) Assertion and reason both are true and the reason is not correct explanation of assertion.
- (2) Assertion and reason both are true but reason in correct explanation of assertion.
- (3) Assertion is true but reason is wrong.
- (4) Assertion and reason both are wrong.

125. The members of following phylum represent tissue level of organization-

- (1) Cnidaria
- (2) Porifera
- (3) Protozoa
- (4) Both (1) and (2)

126. Mark the incorrect statement for the phyla, Platyhelminthes to Echinodermata.

- (1) Groups represent organ/organ - system level of organization
- (2) All are triploblastic
- (3) All are bilaterally symmetrical (except adult echinodermata)
- (4) None of these

127. When any plane passing through the central axis of the body divides the organism into two identical halves, the organism is called-

- (1) Radially symmetrical
- (2) Bilaterally symmetrical
- (3) Asymmetrical
- (4) Metamerically segmented

128. Which of the following is a fresh water sponge?

- (1) *Sycon*
- (2) *Euspongia*
- (3) *Spongilla*
- (4) Frog

129. Comb jellies belong to-

- (1) Porifera
- (2) Cnidaria
- (3) Ctenophora
- (4) Corals

130. The phenomenon of metagenesis occur in-

- (1) *Taenia*
- (2) *Bombyx*
- (3) *Obelia*
- (4) *Musca*

131. Assertion : All porifera are not living in marine water

Reason : *Spongilla* is a fresh water organism

- (1) Assertion and reason both are true and the reason is correct explanation of assertion.
- (2) Assertion and reason both are true but reason in not correct explanation of assertion.
- (3) Assertion is true but reason is wrong.
- (4) Assertion and reason both are wrong.

132. Assertion : Arthropoda is the largest phylum of animals which includes insects.

Reason : Over two-thirds of all named species on earth are arthropoda

- (1) Assertion and reason both are true and the reason is correct explanation of assertion.
- (2) Assertion and reason both are true and the reason is not correct explanation of assertion.
- (3) Assertion is true but reason is wrong.
- (4) Assertion and reason both are wrong.

133. Assertion : Platyhelminthes have dorso-ventrally flattened body, hence are called roundworms.

Reason : Some members like planaria possess low regeneration capacity

- (1) Assertion and reason both are true and the reason is correct explanation of assertion.
- (2) Assertion and reason both are true and the reason is not correct explanation of assertion.
- (3) Assertion is true but reason is wrong.
- (4) Assertion and reason both are wrong.

134. Acoelomate animal is.

- (1) Platyhelminthes
- (2) Annelida
- (3) Mollusca
- (4) Chordata

135. Air bladder is present in:

- (1) Chondrichthyes
- (2) Osteichthyes
- (3) Reptilia
- (4) All fishes

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PLAN A

MLA + MLB + MAP + MAQ + MAS + MPA + SMAR (303031 + 303033 + 303051 + 303054 + 303056 + 303076 + 303082)

DATE	DAY	SYLLABUS	DATE	DAY	SYLLABUS
26-02-2024	Monday	SYLLABUS 1	08-04-2024	Monday	SYLLABUS 5 + 6
03-03-2024	Sunday	SYLLABUS 2	14-04-2024	Sunday	AIOT
09-03-2024	Saturday	SYLLABUS 3	18-04-2024	Thursday	FULL SYLLABUS
15-03-2024	Friday	SYLLABUS 4	21-04-2024	Sunday	AIOT
21-03-2024	Thursday	SYLLABUS 5	24-04-2024	Wednesday	FULL SYLLABUS
27-03-2024	Wednesday	SYLLABUS 6	28-04-2024	Sunday	FULL SYLLABUS
31-03-2024	Sunday	SYLLABUS 1 + 2	02-05-2024	Thursday	FULL SYLLABUS
04-04-2024	Thursday	SYLLABUS 3 + 4			

Proposed Syllabus 1 to 6 (Session 2023-2024)

SUBJECT ➡	PHYSICS	CHEMISTRY	BIOLOGY
SYLLABUS 1	» Physics And Measurement, » Kinematics, » Laws Of Motion, » Work, Energy, And Power » <u>Experimental Skills</u> * Vernier Calipers-Its Use To Measure The Internal And External Diameter And Depth Of A Vessel. * Screw Gauge-Its Use To Determine Thickness/Diameter Of Thin Sheet/Wire.	» Some Basic Concepts of Chemistry, » Redox reactions, » Equilibrium, » Chemical Thermodynamics	» The Living World » Biological Classification » Plant Kingdom » Structural Organisation In Animals (Animal Tissue), » Animal Kingdom » Cockroach, Frog
SYLLABUS 2	» Center Of Mass And Collision, » Circular Motion, » Rotational Motion, » Gravitation, » Properties Of Solids And Liquids, » <u>Experimental Skills</u> * Young'S Modulus Of Elasticity Of The Material Of A Metallic Wire. * Surf Ace Tension Of Water By Capillary Rise And Effect Of Detergents, * Co-Efficient Of Viscosity Of A Given Viscous Liquid By Measuring Terminal Velocity Of A Given Spherical Body	» Atomic Structure, » Classification of Elements and Periodicity in Properties, » Chemical Bonding & Molecular Structure.	» Cell : The Unit Of Life » Cell Cycle And Cell Division » Biomolecules, Enzyme » Excretory Products And Their Elimination (Excretory System) » Breathing And Exchange Of Gases (Respiratory System) » Body Fluids And Circulation (Circulatory System)
SYLLABUS 3	» Thermodynamics, » Kinetic Theory Of Gases, » Oscillations And Waves » <u>Experimental Skills</u> * Simple Pendulum-Dissipation Of Energy By Plotting A Graph Between The Square Of Amplitude And Time. * Metre Scale - The Mass Of A Given Object By The Principle Of Moments. * Speed Of Sound In Air At Room Temperature Using A Resonance Tube, * Specific Heat Capacity Of A Given (I) Solid And (II) Liquid By Method Of Mixtures.	» Some Basic Principles of Organic Chemistry, » Hydrocarbons, » Purification and Characterisation of Organic Compounds.	» Photosynthesis in Higher Plants » Plant Growth and Development » Respiration in Plants » Neural Control And Co-ordination (Nervous System) » Chemical Co-ordination And Integration (Endocrine System) » Locomotion And Movement (Muscles, Skeletal System)
SYLLABUS 4	» Electrostatics & Capacitor, » Current Electricity » <u>Experimental Skills</u> * The Resistivity of The Material of A Given Wire Using A Metre Bridge. * The Resistance Of A Given Wire Using Ohm'S Law. * Resistance And Figure Of Merit Of A Galvanometer By Half Deflection Method.	» Chemical Kinetics » Solutions » Electrochemistry » <u>Principles Related to Pratical Chemistry :-</u> The chemistry involved in the titrimetric exercises-Acids, Bases and the use of indicators, oxalic acid vs KMnO ₄ , Mohr's salt vs KMnO ₄ . Chemical principles involved in the following experiments. 1. Enthalpy of solution of CuSO ₄ . 2. Enthalpy of neutralization of strong acid and strong base. 3. Preparation of lyophilic and lyophobic sols. 4. Kinetic study of the reaction of iodide ions with hydrogen Peroxide at room temperature.	» Sexual Reproduction in Flowering Plants » Morphology Of Flowering Plants » Anatomy Of Flowering Plants » Human Reproduction » Reproductive Health,
SYLLABUS 5	» Magnetic Effects Of Current And Magnetism, » Electromagnetic Induction And Alternating Currents, » Electromagnetic Waves	» p-block elements, » d & f-block elements » Coordination Compounds » <u>Principles Related to Pratical Chemistry :-</u> 1. The chemistry involved in the preparation of Inorganic compound; Mohr's salt, potash alum. 2. Chemical principles involved in the qualitative salt analysis. 3. Cation – Pb ²⁺ Cu ⁺² Al ³⁺ Fe ³⁺ , Zn ⁺² , Ni ²⁺ Ca ²⁺ , Ba ²⁺ , Mg ²⁺ , NH ₄ ⁺ 4. Anions – CO ₃ ²⁻ , S ²⁻ , SO ₄ ²⁻ , NO ₂ ⁻ , Cl ⁻ , Br ⁻ , I ⁻ (Insoluble Salt Excluded)	» Principles Of Inheritance And Variation » Molecular Basis Of Inheritance, Microbes In Human Welfare » Biotechnology : Principles And Processes » Biotechnology And Its Applications
SYLLABUS 6	» Optics, » Dual Nature Of Matter And Radiation, » Atoms And Nuclei, » Electronic Devices » <u>Experimental Skills</u> * The Focal Length Of; (I) Convex Mirror (Ii) Concave Mirror (iii) Convex Lens, Using The Parallax Method. * The Plot Of The Angle Of Deviation Vs Angle Of Incidence For A Triangular Prism. * Refractive Index Of A Glass Slab Using A Travelling Microscope. * Characteristic Curves Of A P-N Junction Diode In Forward And Reverse Bias. * Characteristic Curves Of A Zener Diode And Finding Reverse Break Down Voltage. * Identification Of Diode. Led., Resistor. A Capacitor From A Mixed Collection Of Such Items.	» Organic Compounds Containing Halogens, » Organic Compounds Containing Oxygen, » Organic Compounds Containing Nitrogen, Biomolecules » <u>Principles Related to Pratical Chemistry :-</u> 1. Detection of extra elements (Nitrogen, Sulphur, halogens) in organic compounds. 2. Detection of the hydroxyl (alcoholic and phenolic), carbonyl (aldehyde and ketones), carboxyl and amino Functional groups in organic compounds. 3. The chemistry involved in the preparation of Acetanilide, p-nitro acetanilide , aniline yellow and iodoform.	» Organisms and Populations » Ecosystem » Biodiversity and Conservation » Demography » Biology In Human Welfare : Human Health and Disease » Origin And Evolution

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**BIO-CHEM TEST SERIES**

(Academic Session : 2023 - 2024)

NEET(UG)**Bio-Chemistry****15-02-2024**

**PRE-MEDICAL : MEA+MEB+MEC+JMEA+MLA+MLB+MLC+MLD+
JMLA+JMLD+MAP+MAQ+MAS+MPA+SMAR**

इस पुस्तिका में 36 पृष्ठ हैं। This Booklet contains 36 pages.

इस परीक्षा पुस्तिका को जब तक ना खोलें जब तक कहा न जाए।

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इस परीक्षा पुस्तिका के पिछले आवरण पर दिए निर्देशों को ध्यान से पढ़ें।

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- परीक्षा की अवधि 1½ घंटे है एवं परीक्षा पुस्तिका में 135 प्रश्न हैं। प्रत्येक प्रश्न 4 अंक का है। प्रत्येक सही उत्तर के लिए परीक्षार्थी को 4 अंक दिए जाएंगे। प्रत्येक गलत उत्तर के लिए कुल योग में से एक अंक घटाया जाएगा। अधिकतम अंक 540 हैं।
- इस पृष्ठ पर विवरण अंकित करने एवं उत्तर पत्र पर निशान लगाने के लिए केवल नीले/काले बॉल पॉइंट पेन का प्रयोग करें।
- रफ कार्य इस परीक्षा पुस्तिका में निर्धारित स्थान पर ही करें।
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- परीक्षार्थी सुनिश्चित करें कि इस उत्तर पत्र को मोड़ा न जाए एवं उस पर कोई अन्य निशान न लगाएं। परीक्षार्थी अपना फॉर्म नम्बर प्रश्न पुस्तिका/उत्तर पत्र में निर्धारित स्थान के अतिरिक्त अन्यत्र न लिखें।
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Important Instructions :

- On the Answer Sheet, fill in the particulars on Side-1 and Side-2 carefully with blue/black ball point pen only.
- The test is of 1½ hours duration and this Test Booklet contains 135 questions. Each question carries 4 marks. For each correct response, the candidate will get 4 marks. For each incorrect response, one mark will be deducted from the total scores. The maximum marks are 540.
- Use Blue/Black Ball Point Pen only for writing particulars on this page/markings responses.
- Rough work is to be done on the space provided for this purpose in the Test Booklet only.
- On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
- The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/ Answer Sheet.
- Use of white fluid for correction is not permissible on the Answer Sheet.

प्रश्नों के अनुवाद में किसी अस्पष्टता की स्थिति में, अंग्रेजी संस्करण को ही अंतिम माना जाएगा।

In case of any ambiguity in translation of any question, English version shall be treated as final.

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ENTHUSIAST, LEADER & ACHIEVER COURSE (ALL PHASE)

BIO-CHEMISTRY TEST	DATE : 15 - 02 - 2024
SYLLABUS – 01	
CHEMISTRY : Chemical Equilibrium & Ionic Equilibrium	
BIOLOGY : The Living World, Biological Classification, Plant Kingdom, Animal Kingdom	

- What is wrong about equilibrium state?
 - (1) $\Delta G = 0$
 - (2) The reaction ceases at equilibrium
 - (3) Equilibrium constant is independent of initial concentrations of reactants
 - (4) Catalyst has no effect on equilibrium state.
- The equilibrium constant for the reaction $\text{NH}_4\text{NO}_2(\text{s}) \rightleftharpoons \text{N}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$, is given by
 - (1) $\frac{[\text{NH}_4\text{NO}_2]}{[\text{N}_2][\text{H}_2\text{O}]^2}$
 - (2) $[\text{N}_2][\text{H}_2\text{O}]^2$
 - (3) $\frac{[\text{N}_2][\text{H}_2\text{O}]}{[\text{NH}_4\text{NO}_2]}$
 - (4) $\frac{[\text{N}_2][2\text{H}_2\text{O}]^2}{[\text{NH}_4\text{NO}_2]^2}$
- The equilibrium constant for a reaction $\text{A} + 2\text{B} \rightleftharpoons 2\text{C}$ is 40 then equilibrium constant for reaction $\text{C} \rightleftharpoons \text{B} + \frac{1}{2}\text{A}$ will be :
 - (1) 1/40
 - (2) $(1/40)^{1/2}$
 - (3) $(1/40)^2$
 - (4) 40
- The equilibrium constant (K) for the reaction: $2\text{HI}(\text{g}) \rightleftharpoons \text{H}_2(\text{g}) + \text{I}_2(\text{g})$ at room temperature is 2.85 and that of 698 K is 1.4×10^{-2} . This implies that the forward reaction is
 - (1) Exothermic
 - (2) Endothermic
 - (3) Exergonic
 - (4) Unpredictable

- साम्यावस्था के बारे में क्या गलत है?
 - (1) $\Delta G = 0$
 - (2) अभिक्रिया साम्यावस्था पर रूक जाती है।
 - (3) साम्यावस्था स्थिरांक अभिकारकों की प्रारंभिक सांद्रता से स्वतंत्र होता है।
 - (4) उत्प्रेरक का साम्यावस्था पर कोई प्रभाव नहीं पड़ता।
- अभिक्रिया $\text{NH}_4\text{NO}_2(\text{s}) \rightleftharpoons \text{N}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$ के लिए साम्य स्थिरांक होगा
 - (1) $\frac{[\text{NH}_4\text{NO}_2]}{[\text{N}_2][\text{H}_2\text{O}]^2}$
 - (2) $[\text{N}_2][\text{H}_2\text{O}]^2$
 - (3) $\frac{[\text{N}_2][\text{H}_2\text{O}]}{[\text{NH}_4\text{NO}_2]}$
 - (4) $\frac{[\text{N}_2][2\text{H}_2\text{O}]^2}{[\text{NH}_4\text{NO}_2]^2}$
- अभिक्रिया $\text{A} + 2\text{B} \rightleftharpoons 2\text{C}$ के लिए साम्य स्थिरांक 40 है। अभिक्रिया $\text{C} \rightleftharpoons \text{B} + \frac{1}{2}\text{A}$ के लिए साम्य स्थिरांक होगा
 - (1) 1/40
 - (2) $(1/40)^{1/2}$
 - (3) $(1/40)^2$
 - (4) 40
- अभिक्रिया $2\text{HI}(\text{g}) \rightleftharpoons \text{H}_2(\text{g}) + \text{I}_2(\text{g})$ के लिए साम्य स्थिरांक (K), कमरे के तापमान पर 2.85 है और 698 K पर 1.4×10^{-2} है। इसका तात्पर्य यह है कि अग्र अभिक्रिया होगी
 - (1) ऊष्माक्षेपी
 - (2) ऊष्माशोषी
 - (3) एक्सर्जोनिक
 - (4) नहीं बता सकते हैं

5. For the reaction : $2A(g) + B(g) \rightleftharpoons 3C(g) + 4D(g)$
Two moles each of A and B were taken into a 1L flask. The following must always be true when the system attained equilibrium
(1) $[A] = [B]$
(2) $[A] < [B]$
(3) $[B] = [C]$
(4) $[A] + [B] < [C] + [D]$.
6. In which of the following reaction K_p and K_c are equal
(1) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
(2) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$
(3) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$
(4) $2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$
7. For a hypothetical reaction : $AB_2(g) + \frac{1}{2} B_2(g) \rightleftharpoons AB_3(g)$; $\Delta H = -x$ kJ. More AB_3 could be produced at equilibrium by :
(1) using a catalyst
(2) removing some of B_2
(3) increasing the temperature
(4) increasing the pressure.
8. The rate constants of forward and backward reactions are 8.5×10^{-5} and 2.38×10^{-4} respectively. The equilibrium constant is
(1) 0.35
(2) 0.42
(3) 12.92
(4) 0.292
9. The vapour density of N_2O_4 at a certain temperature is 30. What is the percentage dissociation of N_2O_4 at this temperature?
(1) 53.3%
(2) 106.6%
(3) 26.7%
(4) None
5. अभिक्रिया : $2A(g) + B(g) \rightleftharpoons 3C(g) + 4D(g)$ में A और B के दो-दो मोल 1L फ्लास्क में लिए गए। जब निकाय साम्य अवस्था प्राप्त कर लेगा तो निम्नलिखित हमेशा सत्य होगा -
(1) $[A] = [B]$
(2) $[A] < [B]$
(3) $[B] = [C]$
(4) $[A] + [B] < [C] + [D]$.
6. निम्नलिखित में से किस अभिक्रिया में K_p और K_c बराबर हैं
(1) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
(2) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$
(3) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$
(4) $2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$
7. $AB_2(g) + \frac{1}{2} B_2(g) \rightleftharpoons AB_3(g)$ प्रकार की एक काल्पनिक अभिक्रिया के लिए $\Delta H = -x$ kJ है तो साम्य अवस्था पर अधिक AB_3 का उत्पादन किया जा सकता है
(1) उत्प्रेरक का उपयोग करके
(2) कुछ B_2 को हटा के
(3) तापमान बढ़ाके
(4) दाब बढ़ाके
8. अग्र और पश्च अभिक्रियाओं के दर स्थिरांक क्रमशः 8.5×10^{-5} और 2.38×10^{-4} हैं तो साम्य स्थिरांक है
(1) 0.35
(2) 0.42
(3) 12.92
(4) 0.292
9. एक निश्चित तापमान पर N_2O_4 का वाष्प घनत्व 30 है तो इस तापमान पर N_2O_4 का प्रतिशत वियोजन होगा ?
(1) 53.3%
(2) 106.6%
(3) 26.7%
(4) कोई नहीं।

10. For the following equilibrium
 $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$
 K_p is found to be equal to K_c . This is attained when:
 (1) $T = 1 \text{ K}$
 (2) $T = 12.18 \text{ K}$
 (3) $T = 27.3 \text{ K}$
 (4) $T = 273 \text{ K}$
11. 1 mole each of $\text{CO}(\text{g})$, $\text{H}_2\text{O}(\text{g})$, $\text{H}_2(\text{g})$ and $\text{CO}_2(\text{g})$ are placed in one litre flask at 25°C . When following equilibrium is set-up
 $\text{CO}(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{CO}_2(\text{g}) + \text{H}_2(\text{g}); K_p = 9$.
 Hence CO_2 present at equilibrium is:
 (1) 0.5 mol
 (2) 1.5 mol
 (3) 0.25 mol
 (4) 3.0 mol.
12. One mole of $\text{N}_2\text{O}_4(\text{g})$ is kept in a closed container under one atmosphere at 300 K. When it is heated to 600 K then 20% of $\text{N}_2\text{O}_4(\text{g})$ is converted to $\text{NO}_2(\text{g})$ according to given reaction
 $\text{N}_2\text{O}_4 \rightleftharpoons 2\text{NO}_2(\text{g})$
 then resultant pressure at equilibrium is:
 (1) 1.2 atm
 (2) 2.4 atm
 (3) 2.0 atm
 (4) 1.0 atm.
13. For reaction : $\text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g}) \rightleftharpoons \text{PCl}_5(\text{g})$
 the value of K_c at 250°C is 26 then value of K_p at this temperature will be
 (1) 0.6 (2) 5.7 (3) 8.3 (4) 4.6
14. If K_p for a reaction: $\text{A}(\text{g}) + 2\text{B}(\text{g}) \rightleftharpoons 3\text{C}(\text{g}) + \text{D}(\text{g})$ is 0.05 atm at 1000 K then its K_c in terms of R will be
 (1) 20000 R
 (2) 0.02 R
 (3) $5 \times 10^{-5} \text{ R}$
 (4) $\frac{5 \times 10^{-5}}{\text{R}}$
10. निम्नलिखित साम्य के लिए
 $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$
 K_p तथा K_c बराबर पाया गया। यह सम्भव है यदि :
 (1) $T = 1 \text{ K}$
 (2) $T = 12.18 \text{ K}$
 (3) $T = 27.3 \text{ K}$
 (4) $T = 273 \text{ K}$
11. एक लीटर के फ्लास्क में $\text{CO}(\text{g})$, $\text{H}_2\text{O}(\text{g})$, $\text{H}_2(\text{g})$ और $\text{CO}_2(\text{g})$ प्रत्येक का 1 मोल 25°C तापमान पर लिया गया। निम्न साम्य स्थापित होने पर
 $\text{CO}(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{CO}_2(\text{g}) + \text{H}_2(\text{g}); K_p = 9$.
 साम्य अवस्था पर CO_2 की मात्रा होगी :
 (1) 0.5 मोल
 (2) 1.5 मोल
 (3) 0.25 मोल
 (4) 3.0 मोल
12. 300 K पर $\text{N}_2\text{O}_4(\text{g})$ के एक मोल को एक वायुमंडल दाब पर एक बंद पात्र में लिया गया। यदि $\text{N}_2\text{O}_4(\text{g})$ को 600 K तक गर्म किया गया तो यह निम्न अभिक्रिया के अनुसार 20% $\text{NO}_2(\text{g})$ में परिवर्तित हो जाता है
 $\text{N}_2\text{O}_4 \rightleftharpoons 2\text{NO}_2(\text{g})$
 तो साम्य अवस्था पर परिणामी दाब होगा :
 (1) 1.2 atm
 (2) 2.4 atm
 (3) 2.0 atm
 (4) 1.0 atm.
13. अभिक्रिया : $\text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g}) \rightleftharpoons \text{PCl}_5(\text{g})$ के लिए 250°C पर K_c का मान 26 है तो इस तापमान पर K_p का मान होगा
 (1) 0.6 (2) 5.7 (3) 8.3 (4) 4.6
14. यदि अभिक्रिया : $\text{A}(\text{g}) + 2\text{B}(\text{g}) \rightleftharpoons 3\text{C}(\text{g}) + \text{D}(\text{g})$ के लिए 1000 K पर K_p का मान 0.05 atm है तो R के पदों में इसका K_c होगा
 (1) 20000 R
 (2) 0.02 R
 (3) $5 \times 10^{-5} \text{ R}$
 (4) $\frac{5 \times 10^{-5}}{\text{R}}$

15. In which of the following cases, the reaction goes near to completion?
- $K = 10^3$
 - $K = 10^{-2}$
 - $K = 10$
 - $K = 10^0$.
16. For a reaction: $H_2 + I_2 \rightleftharpoons 2HI$, the value of equilibrium constant is 50 at 271 K. If 0.5 moles each of H_2 and I_2 is added to the reaction system, then value of equilibrium constant will be
- 0.02
 - 0.2
 - 50
 - 25.
17. 1.1 mol of A is mixed with 2.2 mol of B and the mixture is kept in one litre flask till the equilibrium is reached. If 0.2 mol of C is formed at equilibrium then value of equilibrium constant for the given reaction $A + 2B \rightarrow 2C + D$ is
- 0.002
 - 0.004
 - 0.001
 - 0.003.
18. In which of the following reaction $K_p > K_c$?
- $N_2(g) + 2H_2(g) \rightleftharpoons 2NH_3(g)$
 - $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$
 - $PCl_3(g) + Cl_2 \rightleftharpoons PCl_5(g)$
 - $2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$.
19. For the reaction : $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$, the forward reaction at constant temperature is favoured by :
- introduction of an inert gas at constant volume
 - introduction of $PCl_3(g)$ at constant volume
 - introduction of $PCl_5(g)$ at constant volume
 - introduction of Cl_2 at constant volume.
15. निम्नलिखित में से किस विकल्प में अभिक्रिया पूर्णता के सबसे अधिक निकट है?
- $K = 10^3$
 - $K = 10^{-2}$
 - $K = 10$
 - $K = 10^0$.
16. निम्न अभिक्रिया $H_2 + I_2 \rightleftharpoons 2HI$ के साम्य स्थिरांक का मान 271 K पर 50 है। यदि अभिक्रिया निकाय में H_2 और I_2 प्रत्येक का 0.5 मोल मिला दिया जाये तो साम्य स्थिरांक का मान होगा
- 0.02
 - 0.2
 - 50
 - 25
17. जब A के 1.1 मोल को B के 2.2 मोल के साथ मिलाया जाता है और मिश्रण को साम्य अवस्था पर पहुँचने तक एक लीटर के फ्लास्क में रखा जाता है। यदि साम्य अवस्था पर C का 0.2 mol बनता है तो अभिक्रिया $A + 2B \rightarrow 2C + D$ का साम्य स्थिरांक होगा
- 0.002
 - 0.004
 - 0.001
 - 0.003.
18. निम्नलिखित में से किस अभिक्रिया में $K_p > K_c$ है?
- $N_2(g) + 2H_2(g) \rightleftharpoons 2NH_3(g)$
 - $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$
 - $PCl_3(g) + Cl_2 \rightleftharpoons PCl_5(g)$
 - $2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$.
19. अभिक्रिया: $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ के लिए स्थिर तापमान पर अग्र अभिक्रिया के लिए अनुकूल परिस्थिति होगी
- स्थिर आयतन पर अक्रिय गैस का योग
 - स्थिर आयतन पर $PCl_3(g)$ का योग
 - स्थिर आयतन पर $PCl_5(g)$ का योग
 - स्थिर आयतन पर Cl_2 का योग

20. In the case of gaseous homogeneous reaction, the active mass of the reactant is obtained by the expression

(1) $\frac{PV}{RT}$

(2) $\frac{P}{RT}$

(3) $\frac{RT}{P}$

(4) $\frac{n}{V}RT$

21. The chemical equilibrium of a reversible reaction is not influenced by

(1) pressure

(2) catalyst

(3) concentration of the reactants

(4) temperature.

22. $\text{CaCO}_3 \rightleftharpoons \text{CaO} + \text{CO}_2$ the given reaction goes to completion in a lime kiln because

(1) CaO does not react with CO_2 to give CaCO_3

(2) backward reaction is very slow

(3) CO_2 formed escapes out

(4) None of these

23. The K_{sp} of Ag_2CrO_4 is 1.1×10^{-12} at 298 K then solubility (in mol/L) of Ag_2CrO_4 in 0.1 M AgNO_3 solution will be

(1) 1.1×10^{-11}

(2) 1.1×10^{-10}

(3) 1.1×10^{-12}

(4) 1.1×10^9

24. The Bronsted acid which gives the weakest conjugate base is

(1) HF

(2) H_2S

(3) HCl

(4) H_2O .

20. गैसीय समांगी अभिक्रिया के लिए अभिकारक का सक्रिय द्रव्यमान निम्न प्रकार प्राप्त किया जा सकता है

(1) $\frac{PV}{RT}$

(2) $\frac{P}{RT}$

(3) $\frac{RT}{P}$

(4) $\frac{n}{V}RT$

21. किसी उत्क्रमणीय अभिक्रिया की रासायनिक साम्य अवस्था निम्न से प्रभावित नहीं होती है

(1) दाब

(2) उत्प्रेरक

(3) अभिकारकों की सांद्रता

(4) तापमान

22. चूने के भट्टे में $\text{CaCO}_3 \rightleftharpoons \text{CaO} + \text{CO}_2$ अभिक्रिया पूर्ण हो जाती है क्योंकि

(1) CaO, CO_2 के साथ क्रिया करके CaCO_3 नहीं देता है

(2) पश्च अभिक्रिया बहुत धीमी होती है

(3) बनी हुई CO_2 बाहर निकल जाती है

(4) इनमें से कोई नहीं

23. 298 K पर Ag_2CrO_4 का $K_{sp} = 1.1 \times 10^{-12}$ है तो 0.1 M AgNO_3 के विलयन में Ag_2CrO_4 की विलेयता (mol/L में) होगी

(1) 1.1×10^{-11}

(2) 1.1×10^{-10}

(3) 1.1×10^{-12}

(4) 1.1×10^9

24. निम्न में से कौनसा ब्रॉंस्टेड अम्ल सबसे दुर्बल क्षार बनायेगा

(1) HF

(2) H_2S

(3) HCl

(4) H_2O .

25. Which of the following is a Lewis acid?
 (1) H_2O
 (2) SnCl_4
 (3) $\text{C}_2\text{H}_5\text{OH}$
 (4) Cl^- .
26. NH_4OH is a weak base but it becomes still weaker in the aqueous solution of
 (1) 0.1 M HCl
 (2) 0.1 M NH_4Cl
 (3) 0.1 M H_2SO_4
 (4) 0.1 M CH_3COOH .
27. The addition of NaCl to AgCl decreases the solubility of AgCl because
 (1) Solubility product decreases
 (2) Solubility product remains constant
 (3) Solution becomes unsaturated
 (4) Solution becomes supersaturated.
28. Which pair will show common ion effect?
 (1) $\text{BaCl}_2 + \text{Ba}(\text{NO}_3)_2$
 (2) $\text{NaCl} + \text{HCl}$
 (3) $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$
 (4) $\text{AgCN} + \text{KCN}$.
29. Which of the following has maximum pH?
 (1) 1N CH_3COOH
 (2) 1N HCl
 (3) 1N H_2SO_4
 (4) 1N HNO_3 .
30. The value of pK_w of water
 (1) increases with increase in temperature
 (2) decreases with increase in temperature
 (3) does not change with variation in temperature
 (4) increases till 50°C and there after decreases.
31. When 2 moles of HCl is added to 1 litre of an acidic buffer, its pH changes from 3.4 to 2.9. The buffer capacity of the buffer solution is
 (1) 2 (2) 0
 (3) 4 (4) 8
25. निम्नलिखित में से कौनसा लुईस अम्ल है?
 (1) H_2O
 (2) SnCl_4
 (3) $\text{C}_2\text{H}_5\text{OH}$
 (4) Cl^- .
26. NH_4OH एक दुर्बल क्षार है निम्न में से किसके जलीय विलयन में यह और भी दुर्बल हो जाता है
 (1) 0.1 M HCl
 (2) 0.1 M NH_4Cl
 (3) 0.1 M H_2SO_4
 (4) 0.1 M CH_3COOH .
27. AgCl में NaCl मिलाने से AgCl की विलेयता कम हो जाती है क्योंकि
 (1) विलेयता गुणनफल घट जाता है
 (2) विलेयता गुणनफल स्थिर रहता है
 (3) विलयन असंतृप्त हो जाता है
 (4) विलयन अतिसंतृप्त हो जाता है।
28. निम्न में से कौनसा युग्म समआयन प्रभाव दर्शायेगा ?
 (1) $\text{BaCl}_2 + \text{Ba}(\text{NO}_3)_2$
 (2) $\text{NaCl} + \text{HCl}$
 (3) $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$
 (4) $\text{AgCN} + \text{KCN}$.
29. निम्नलिखित में से किसका pH अधिकतम है?
 (1) 1N CH_3COOH
 (2) 1N HCl
 (3) 1N H_2SO_4
 (4) 1N HNO_3 .
30. पानी के pK_w का मान
 (1) तापमान में वृद्धि के साथ वृद्धि होती है
 (2) तापमान बढ़ने के साथ घटता है
 (3) तापमान में परिवर्तन के साथ परिवर्तित नहीं होता है
 (4) 50°C तक बढ़ता है और उसके बाद घटता है।
31. जब 1 लीटर अम्लीय बफर में 2 मोल HCl मिलाया जाता है, तो इसका pH, 3.4 से 2.9 तक परिवर्तित होता है तो बफर विलयन की बफर क्षमता होगी
 (1) 2 (2) 0
 (3) 4 (4) 8

32. The pH of a dilute solution of acetic acid was found to be 4.3 then addition of a small crystal of sodium acetate will cause pH to
- become less than 4.3
 - become more than 4.3
 - remain equal to 4.3
 - unpredictable.
33. Which of the following is not a buffer solution?
- $\text{CH}_3\text{COOH} + \text{CH}_3\text{COONa}$
 - $\text{HCl} + \text{NaCl}$
 - $\text{HCOOH} + \text{HCOONa}$
 - $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$
34. Which indicator should be used in titrating KOH against oxalic acid?
- Litmus
 - Methyl orange
 - Methyl red
 - Phenolphthalein.
35. Three sparingly soluble salts M_2X , MX and MX_3 have same value of solubility product. Their solubilities follow the order
- $\text{MX}_3 > \text{MX} > \text{M}_2\text{X}$
 - $\text{MX} > \text{MX}_3 > \text{M}_2\text{X}$
 - $\text{MX} > \text{M}_2\text{X} > \text{MX}_3$
 - $\text{MX}_3 > \text{M}_2\text{X} > \text{MX}$.
36. Solubility of a salt A_2B_3 is $1 \times 10^{-4} \text{ mol L}^{-1}$ then its solubility product is
- 1.08×10^{20}
 - 1.08×10^{18}
 - 2.6×10^{-18}
 - 1.08×10^{-18} .
32. एसिटिक अम्ल के तनु विलयन का pH = 4.3 पाया गया तो सोडियम एसिटेट का एक छोटा क्रिस्टल मिलाने से विलयन की pH :
- 4.3 से कम हो जायेगा
 - 4.3 से अधिक हो जायेगा
 - 4.3 के बराबर रहेगा
 - नहीं बता सकते
33. निम्नलिखित में से कौन सा बफर विलयन नहीं है?
- $\text{CH}_3\text{COOH} + \text{CH}_3\text{COONa}$
 - $\text{HCl} + \text{NaCl}$
 - $\text{HCOOH} + \text{HCOONa}$
 - $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$
34. ऑक्जेलिक अम्ल का KOH के साथ अनुमापन करने में किस सूचक का उपयोग किया जाना चाहिए?
- लिटमस
 - मिथाइल ऑरेंज
 - मेथिल रेड
 - फिनोल्फथेलिन
35. तीन अल्प विलेयशील लवणों M_2X , MX और MX_3 का विलेयता गुणनफल का मान समान है तो इनकी विलेयता का क्रम होगा
- $\text{MX}_3 > \text{MX} > \text{M}_2\text{X}$
 - $\text{MX} > \text{MX}_3 > \text{M}_2\text{X}$
 - $\text{MX} > \text{M}_2\text{X} > \text{MX}_3$
 - $\text{MX}_3 > \text{M}_2\text{X} > \text{MX}$.
36. एक लवण A_2B_3 की विलेयता $1 \times 10^{-4} \text{ mol L}^{-1}$ है तो उसका विलेयता गुणनफल है
- 1.08×10^{20}
 - 1.08×10^{18}
 - 2.6×10^{-18}
 - 1.08×10^{-18} .

37. If K_a of HCN = 4×10^{-10} then the pH of 2.5×10^{-1} molar HCN(aq) is
 (1) 2×10^{-2}
 (2) 4.7
 (3) 2.0
 (4) 5.0
38. The pH of 10^{-10} M NaOH solution at 25°C is approximately
 (1) 7
 (2) 10
 (3) between 10 and 11
 (4) between 6 and 7.
39. The dissociation constant of a weak acid HA is 1×10^{-4} then equilibrium constant for its reaction with strong base BOH is
 (1) 1×10^{-4}
 (2) 1×10^{-10}
 (3) ∞
 (4) 1×10^{10}
40. K_a for HCN is 5×10^{-10} at 25°C then how much volume of 5 M KCN solution is required to be added to 10 mL of 2 M HCN in order to maintain pH = 9.
 (1) 2 mL
 (2) 5 mL
 (3) 3 mL
 (4) 4 mL.
41. pK_b of NH_3 is 4.74. Then what will be the pH when 100 mL of 0.01 M NH_3 solution is 50% neutralised by 0.01 M HCl
 (1) 4.74
 (2) 2.37
 (3) 9.26
 (4) 9.48
37. यदि HCN का $K_a = 4 \times 10^{-10}$ है तो 2.5×10^{-1} मोलर HCN(aq) का pH है
 (1) 2×10^{-2}
 (2) 4.7
 (3) 2.0
 (4) 5.0
38. 25°C पर 10^{-10} M NaOH विलयन का pH लगभग होगा -
 (1) 7
 (2) 10
 (3) 10 से 11 के बीच
 (4) 6 से 7 के बीच
39. एक दुर्बल अम्ल HA का वियोजन स्थिरांक 1×10^{-4} है तो इसकी प्रबल क्षार BOH के साथ अभिक्रिया के लिए साम्य स्थिरांक होगा
 (1) 1×10^{-4}
 (2) 1×10^{-10}
 (3) ∞
 (4) 1×10^{10}
40. 25°C पर HCN के लिए $K_a = 5 \times 10^{-10}$ है तो pH = 9 का विलयन बनाए रखने के लिए 2 M HCN के 10 mL में 5 M KCN विलयन की कितनी मात्रा मिलाने की आवश्यकता है।
 (1) 2 mL
 (2) 5 mL
 (3) 3 mL
 (4) 4 mL.
41. NH_3 का $pK_b = 4.74$ है तो 0.01 M NH_3 विलयन के 100 mL का 0.01 M HCl द्वारा 50% उदासीनीकरण होने पर pH होगी
 (1) 4.74
 (2) 2.37
 (3) 9.26
 (4) 9.48

42. The solubility of PbCl_2 is

- (1) $\sqrt{K_{sp}}$
- (2) $(K_{sp})^{1/3}$
- (3) $(K_{sp}/4)^{1/3}$
- (4) $(8K_{sp})^{1/2}$

43. If a neutral solution has $\text{pK}_w = 13.36$ at 50°C , then pH of the solution is

- (1) 6.68
- (2) 7
- (3) 7.68
- (4) None of these.

44. The pH value of M/1000 solution of KOH in water is

- (1) 3
- (2) 6
- (3) 11
- (4) 9

45. **Assertion (A)** : For equilibrium $X_{(g)} \rightleftharpoons Y_{(g)}$

total no. of moles at new equilibrium is less than old equilibrium if some amount of $X_{(g)}$ is removed.

Reason (R) : The no. of moles of the substance removed is partially compensated as the system reaches to new equilibrium.

- (1) Both A and R are true and R is the correct explanation of A.
- (2) Both A and R are true and R is not a correct explanation of A.
- (3) A is true but R is false
- (4) A is false but R is true.

42. PbCl_2 की विलेयता है

- (1) $\sqrt{K_{sp}}$
- (2) $(K_{sp})^{1/3}$
- (3) $(K_{sp}/4)^{1/3}$
- (4) $(8K_{sp})^{1/2}$

43. यदि किसी उदासीन विलयन का 50°C पर $\text{pK}_w = 13.36$ है, तो विलयन की pH होगी

- (1) 6.68
- (2) 7
- (3) 7.68
- (4) इनमें से कोई नहीं

44. जल में KOH के M/1000 विलयन का pH होगा

- (1) 3
- (2) 6
- (3) 11
- (4) 9

45. **कथन (A)** : दिये गये साम्य के लिए $X_{(g)} \rightleftharpoons Y_{(g)}$

यदि $X_{(g)}$ की कुछ मात्रा हटा ली जाए तो नये साम्य पर कुल मोलों की संख्या पुराने साम्य पर कुल मोलों की संख्या से कम होती है।

कारण (R) : जब निकाय नये साम्य पर पहुँचता है तो हटाये गये पदार्थ के मोलों की संख्या आंशिक रूप से कम्पेंसेट हो जाता है।

- (1) दोनों A तथा R सत्य हैं तथा R, A का सही स्पष्टीकरण है
- (2) दोनों A तथा R सत्य हैं तथा R, A का सही स्पष्टीकरण नहीं है
- (3) A सत्य है लेकिन R असत्य है
- (4) A असत्य है लेकिन R सत्य है

46. Polysulphate ester is present in the cell wall of A which also show B. What are A and B respectively:
- (1) A = Green algae, B = A granal chloroplast
 - (2) A = Red algae, B = Isogamy
 - (3) A = Brown algae, B = Heterotrichous thallus
 - (4) A = Red Algae, B = Advanced type of sexual reproduction
47. Which of the following option correctly identifies the plant shown in figure and group it belongs to?



- (1) Marchantia - a liverwort
 - (2) Sphagnum - a moss
 - (3) Sphgnum - a Liverwort
 - (4) Funaria - a moss
48. Red algae differ from the brown algae in having:
- (1) Chlorophyll 'a'
 - (2) Aquatic nature
 - (3) Cellulosic cell wall
 - (4) Floridean starch
49. Pyrenoids are
- (1) Core of starch surrounded by protein
 - (2) Core of protein surrounded by fats
 - (3) Proteinaceous centre and surrounded by starch
 - (4) Core of nucleic acid surrounded by protein
50. Classification of algae is mainly based on:-
- (1) Type of pigments
 - (2) Stored food
 - (3) Type of reproduction
 - (4) Cell wall

46. पॉलीसल्फेट एस्टर A की कोशिका भित्ति में उपस्थित होता है जिसमें B भी पाया जाता है A तथा B क्रमशः
- (1) A = हरी शैवाल, B = A ग्रेना रहित हरितलवक
 - (2) A = लाल शैवाल, B = समयुग्मन
 - (3) A = भूरी शैवाल, B = विषम तन्तुकी थैलस
 - (4) A = लाल शैवाल, B = विकसित प्रकार का लैंगिक जनन
47. निम्न में से कौन सा विकल्प नीचे दिये गये पौधे एवं उससे संबंधित समूह से सुमेलित है?



- (1) मार्केन्शिया - एक लिवरवर्ट
 - (2) स्फैगनम - एक मॉस
 - (3) स्फैगनम - एक लिवरवर्ट
 - (4) फ्यूनेरिया - एक मॉस
48. लाल शैवाल किसकी उपस्थिति में भूरी शैवाल से भिन्न है?
- (1) पर्णहरित 'a' रखने में
 - (2) जलीय प्रकृति रखने में
 - (3) सेल्यूलोजिक कोशिका भित्ति रखने में
 - (4) फ्लोरिडियन स्टार्च में
49. पायरेनॉयड होते हैं-
- (1) प्रोटीन से घिरी हुई स्टार्च की कोर
 - (2) वसा से घिरी हुई प्रोटीन की कोर
 - (3) स्टार्च से घिरी हुई केन्द्रीय प्रोटीन
 - (4) प्रोटीन से घिरा हुआ न्यूक्लिक अम्ल का कोर
50. शैवाल का वर्गीकरण मुख्यतः आधारित होता है
- (1) वर्णकों के प्रकार पर
 - (2) संचित भोजन पर
 - (3) प्रजनन प्रकार पर
 - (4) कोशिका भित्ति पर

51. Which of the following statement is correct?

- (1) Brown algae possess chlorophyll a, b carotenoids, and xanthophyll.
- (2) Majority of red algae are found in fresh water
- (3) Volvox, Fucus and Ulothrix are green algae
- (4) Sexual Reproduction may be isogamous, anisogamous or oogamous in brown algae.

52. Antherozoid of bryophytes are usually:-

- (1) Non-flagellated coma shaped
- (2) Biflagellated, coma shaped
- (3) Multiflagellated, spiral
- (4) Monoflagellated, spiral

53. Select the incorrect statement from the followings:-

- (1) The plant Body of liverworts is not thalloid but made up of rhizoids, stem and leaves.
- (2) Mosses have upright, slender axis bearing spirally arranged leaves like structure
- (3) Spores germinate to form gametophyte in liverwort
- (4) The zygote produces a sporophyte in embryophyta

54. Which one of following statement regarding bryophytes is correct?

- (1) The plant body is less differentiated than that of algae
- (2) They have true roots and leaves but not stem
- (3) Antherozooids are mainly multiciliated
- (4) Zyotes do not undergo reduction division immediately

51. निम्न में से कौनसा कथन सही है?

- (1) भूरी शैवाल में, पर्णहरिम a, b कैरोटिनोइड और जेम्थोफिल उपस्थित होते हैं।
- (2) अधिकतर लाल शैवाल स्वच्छ जल में पाये जाते हैं।
- (3) वॉलवाक्स, फ्युकस तथा यूलोथ्रिक्स हरी शैवाल हैं।
- (4) भूरी शैवाल में लैंगिक जनन समयुग्मकी, असमयुग्मकी या अण्डयुग्मकी हो सकता है।

52. ब्रायोफाइट्स के एन्थ्रोजाइड (नर युग्मक) सामान्यतः होते हैं:-

- (1) अकाशाभिक, कोमाकार
- (2) द्विकशाभिक, कोमा आकार
- (3) बहुकशाभिक, सर्पिलाकार
- (4) एक कशाभिक, सर्पिलाकार

53. निम्नलिखित में से गलत कथन चुनिए:-

- (1) लिवरवर्ट का पादप काय थैलोइड नहीं होता लेकिन मूलाभास, तना व पतियों का बना होता है।
- (2) मॉस में सर्पिलाकार पर्ण समान संरचना युक्त उर्ध्व पतला अक्ष होता है
- (3) लीवरवर्ट में बीजाणु अंकुरित होकर युग्मकोद्भिद बनाते हैं।
- (4) भ्रूणीय पादपो में युग्मनज से बीजाणुद्भिद बनते हैं।

54. ब्रायोफाइट के संदर्भ में निम्न में से कौनसा कथन सत्य है?

- (1) इनकी पादप काय शैवाल की अपेक्षा कम विभेदित होती है।
- (2) इनमें वास्तविक मूल व पतियाँ होती हैं परन्तु वास्तविक तना नहीं होता
- (3) पुमणु मुख्यतः बहुपक्षाभिक होते हैं
- (4) युग्मनज में तुरंत न्यूनीकरण विभाजन (मिऑसिस) नहीं होता है

55. Read the following statements:-

- (i) The predominant stage of life cycle is the gametophyte
- (ii) The sex organs antheridia and archegonia are produced at the apices of the leafy shoot
- (iii) Vegetative reproduction is by fragmentation and budding in the secondary protonema

Which of the following is correctly matched with all the above given statement (i-iii)?

Option:-

- (1) Riccia
- (2) Selaginella
- (3) Chara
- (4) Funaria

56. Bryophytes resemble algae in:-

- (1) Differentiation of the plant body into root, stem and heterotrophic mode of nutrition
- (2) Gametophytic like plant body, lack of vascular tissue, absence of roots and autotrophic mode of nutrition
- (3) Thallus like plant body, presence of roots and heterotrophic mode of nutrition
- (4) Filamentous body, presence of vascular tissue and autotrophic mode of nutrition.

57. What is common among Porphyra, Agaricus, Chlorella and Spirulina?

- (1) They all are photosynthetic
- (2) They all have eukaryotic cell organization
- (3) They all are single celled
- (4) They all either used as food or food supplements

55. निम्न कथनों को पढ़िये-

- (i) जीवन चक्र की प्रभावी अवस्था युग्मकोद्भिद् होती है।
- (ii) लैंगिक अंग पुंधानी तथा स्त्रीधानी पर्णिल प्ररोह के शीर्षों पर स्थित होते हैं।
- (iii) कायिक जनन खण्डन के द्वारा तथा द्वितीयक प्रोटोनीमा में मुकुलन के द्वारा होता है

निम्न में से कौनसे एक विकल्प के लिए उपरोक्त सभी दिये गये सही है?

विकल्प:-

- (1) रिक्सिया
- (2) सिलेजिनेला
- (3) कारा
- (4) फ्यूनेरिया

56. ब्रायोफाइट्स शैवाल से समानता रखते हैं:-

- (1) पादप शरीर का जड़, तने में विभेदित होना तथा विषमपोषी पोषण से
- (2) युग्मकोद्भिद पादप शरीर, संवहन उत्तक की अनुपस्थिति, जड़ की अनुपस्थिति तथा स्वपोषी पोषण
- (3) थैलस के समान पादप शरीर, जड़ की उपस्थिति तथा विषमपोषी पोषण
- (4) तन्तुमय शरीर, संवहन उत्तक की उपस्थिति तथा स्वपोषित पोषण

57. पोरफायरा, अगेरीकस, क्लोरेला एवं स्पाइरुलिना में क्या समान है?

- (1) ये सभी प्रकाश सश्लेषी हैं
- (2) इन सभी में यूकेरियोटिक कोशिकीय संगठन होता है
- (3) ये सभी एकल कोशिकीय होते हैं
- (4) ये सभी भोजन या भोजन सम्पूरक के रूप में उपयोग में लिए जाते हैं

58. Match the following Column-I and Column-II :-

	Column-I		Column-II
A	Ulothrix	i	Isogamy
B	Fucus	ii	Diplontic
C	Polysiphonia	iii	Oogamy

- (1) A-i, B-ii, C-iii (2) A-i, B-iii, C-ii
(3) A-ii, B-iii, C-i (4) A-iii, C-ii, C-i

59. Which one of the following statements is true?

- (1) Agar-agar is obtained from Gelidium and Gracilaria
(2) Laminaria and Sargassum are used as food
(3) Algae increase the level of dissolved oxygen in the immediate environment
(4) All of the above

60. Read the following statements carefully and identify two correct statement:-

- (A) In brown algae fusion of gametes occurs either in water or inside oogonium
(B) Pigments of blue green algae are similar to brown algae
(C) Chlorophyll a & d are present in red algae
(D) Haplontic life cycle is observed in algae & bryophyta

Option:-

- (1) B and C (2) A and B
(3) A and C (4) A and D

61. Which of the following statement is incorrect for Algae?

- (1) Two, unequal lateral flagella are found in brown algae
(2) Chlorophyll-a is present in green, brown and red algae
(3) The red algae usually reproduce vegetatively by fragmentation
(4) Zoospores are found in red algae

58. कॉलम-I और कॉलम-II का मिलान कीजिए:-

	कॉलम-I		कॉलम-II
A	युलोथ्रिक्स	i	समयुग्मन
B	फ्यूकस	ii	द्विगुणितक
C	पोलीसाइफोनिया	iii	अण्डयुग्मन

- (1) A-i, B-ii, C-iii (2) A-i, B-iii, C-ii
(3) A-ii, B-iii, C-i (4) A-iii, C-ii, C-i

59. निम्नलिखित में से कौनसा कथन सत्य है?

- (1) ऐगार-ऐगार, जेलीडियम और ग्रेसिलेरिया से प्राप्त किया जाता है।
(2) लेमिनेरिया और सारगोसम खाद्य के रूप में प्रयुक्त किये जाते हैं।
(3) शैवाल अपने सन्निकट पर्यावरण में घुलित ऑक्सीजन के स्तर को बढ़ाती हैं।
(4) उपरोक्त सभी

60. निम्न कथनों का ध्यानपूर्वक पढ़िये तथा दो सही कथनों का पहचानिये:-

- (A) भूरे शैवाल में युग्मको का संलयन जल में या अण्डधानी के अन्दर होता है
(B) नील हरित शैवाल के वर्णन भूरे शैवाल के समान होता है
(C) लाल शैवाल में पर्णहरिम a तथा d पाया जाता है
(D) शैवाल तथा ब्रायोफायटा में हैप्लोण्टिक जीवन चक्र पाया जाता है

Option:-

- (1) B तथा C (2) A तथा B
(3) A तथा C (4) A तथा D

61. शैवालों के संदर्भ में निम्नलिखित में कौनसा कथन गलत है?

- (1) भूरे शैवालों में दो, असमान पार्श्व कशाभिक पाये जाते हैं।
(2) हरे, भूरे तथा शैवालों में क्लोरोफिल a पाया जाता है।
(3) लाल शैवाल प्रायः खण्डन द्वारा कायिक प्रजनन करते हैं
(4) लाल शैवालों में चलबीजाणु (Zoospores) पाये जाते हैं।

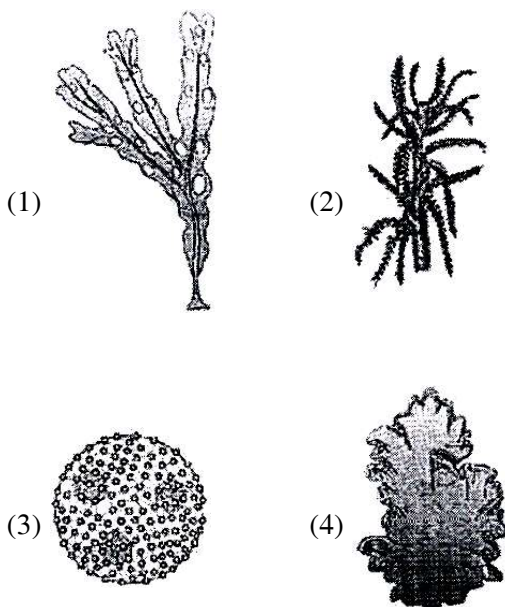
62. Mosses are ecologically important as:-

- (1) They are source of absorbant cotton
- (2) They reduces soil erosion
- (3) Mosses are first organism to colonise water lake
- (4) All of the above

63. Which of the following statement is correct?

- (1) In mosses asexual reproduction takes place by green, unicellular buds called gammae
- (2) The sporophyte of moss is differentiated in foot, seta and capsule
- (3) Protonema is a green, creeping, branched and filamentous stage in red algae, which develops directly from gametes.
- (4) Moss have an elaborate mechanism of seed dispresal.

64. In the following diagrams which one provides algin that have long been used in ice cream?



65. Which of the following algae show diplontic life cycle?

- (1) Ulothrix
- (2) Spirogyra
- (3) Ectocarpus
- (4) Fucus

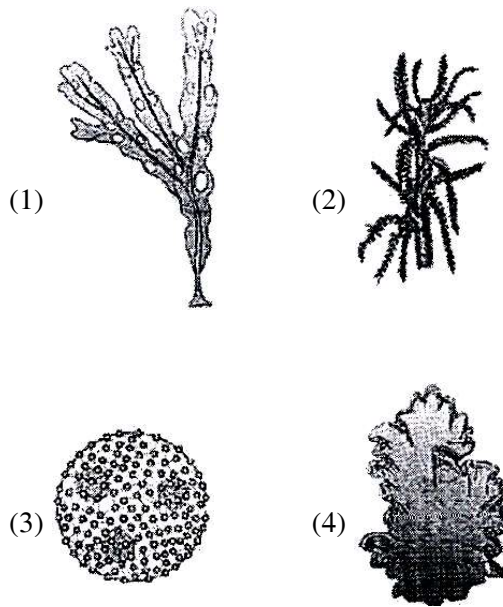
62. मॉसेज पारिस्थितिकीय रूप से महत्वपूर्ण है क्योंकि

- (1) यह अवशोषी कपास के स्रोत है
- (2) यह मृदा अपरदन को कम करते है
- (3) मॉसेज जलीय झील में उगने वाले प्रथम जीव होते है
- (4) उपरोक्त सभी

63. निम्न में से कौनसा कथन सही है ?

- (1) मॉस में, अलैंगिक जनन हरी, बहुकोशिकीय कलियों द्वारा होता है, जिन्हे गेमी कहा जाता है।
- (2) मॉस का बीजाणुद्विद पाद, सीटा और कैप्सूल में विभेदित होता हैं
- (3) प्रोटोनीमा (प्राकतंतु) लाल शैवाल की एक हरी, विसर्पी शाखित और तंतुमय अवस्था है, जो की सीधे युग्मको से विकसित होता है।
- (4) मॉस में बीज प्रकीर्णन की विस्तृत प्रणाली होती है।

64. निम्नलिखित चित्रों में से कौनसा एक एल्जिन प्रदान करता है जो लम्बे समय आइसक्रीम में उपयोग में आ रहा है ?



65. निम्नलिखित में से कौनसा शैवाल द्विगुणितक जीवन चक्र दर्शाता है ?

- (1) यूलोथ्रिक्स
- (2) स्पाइरोगायरा
- (3) एक्टोकार्पस
- (4) फ्युकस

66. Find the correct about algae
- (1) They have vascular tissues
 - (2) They produce antheridia as female sex organ
 - (3) They produce archegonia as female sex organ
 - (4) They produce oogonia as female sex organ/structure
67. Which of the following pair is an example of heterosporous pteridophyte ?
- (1) *Psilotum* and *Adiantum*
 - (2) *Salvinia* and *Selaginella*
 - (3) *Lycopodium* and *Dryopteris*
 - (4) *Pteris* and *Adiantum*
68. In pteridophyte gametophyte (prothallus) prefers to grow in moist shady places because :
- (1) In these conditions, it survive very well
 - (2) It represents the ancestry of bryophytes
 - (3) It need water for fertilization
 - (4) In pteridophytes roots are short
69. Consider the following four statements (a-d) and select the option which include the all correct ones only :-
- (a) Unlike bryophytes and pteridophytes, in gymnosperms the male and female gametophyte do not have an independent free living existence.
 - (b) In pteridophytes, the female gametophyte is retained on the parent sporophyte for variable period.
 - (c) Zygote is the first cell of sporophyte generation
 - (d) Bryophytes do not play any role in plant succession occurring on bare rock or soil or Terrestrial habitate.
- (1) b, c and d
 - (2) a, c and d
 - (3) a, b and d
 - (4) a, b and c
66. शैवालों के बारे में सत्य को पहचानिये
- (1) इनमें संवहन ऊतक होते हैं
 - (2) इसमें मादा जननांग के रूप में पुंधानियाँ बनती हैं
 - (3) इनमें मादा जननांग के रूप में स्त्रीधानियाँ बनती हैं
 - (4) इनमें मादा जननांग/संरचनाएँ अण्डधानियाँ के रूप में बनती हैं
67. निम्नलिखित में से कौनसा जोड़ा विषमबीजाणुक टेरिडोफाइट का उदाहरण है ?
- (1) साइलोटम एवं एडिऐन्टम
 - (2) साल्विनिया एवं सिलैजिनेला
 - (3) लाइकोपोडियम एवं ड्रायोप्टैरिस
 - (4) टेरिस एवं एडिऐन्टम
68. टेरिडोफाइट में युग्मकोद्भिद (प्रोथेलस) नमी छायादार वाले स्थानों पर उगना पसन्द करता है क्योंकि :-
- (1) इस प्रकार की परिस्थितियों में ये अच्छी प्रकार से पनपता है।
 - (2) यह ब्रायोफाइट के साथ पूर्वजता को प्रदर्शित करता है।
 - (3) इसको निषेचन के लिए जल की आवश्यकता होती है।
 - (4) टेरिडोफाइट में जड़े छोटी होती हैं।
69. नीचे दिये गये चार कथनों (a-d) को पढ़िये तथा केवल सही कथनों वाले विकल्प को चुनिये :-
- (a) ब्रायोफाइट्स तथा टेरिडोफाइट्स के विपरीत जिम्नोस्पर्म में नर तथा मादा युग्मकोद्भिद स्वतंत्रजीवी के रूप में नहीं पाये जाते हैं।
 - (b) टेरिडोफाइट्स में मादा युग्मोद्भिद बीजाणुद्भिद पर विभिन्न समय के लिए आश्रित रहता है।
 - (c) युग्मनज बीजाणुद्भिद पीढ़ी की प्रथम कोशिका होती है
 - (d) नग्न चट्टानों मृदा अथवा स्थलीय आवासों पर होने वाले अनुक्रमण में ब्रायोफाइट्स कोई भूमिका नहीं निभाते हैं।
- (1) b, c और d
 - (2) a, c और d
 - (3) a, b और d
 - (4) a, b और c

70. Select the true statement from the following :-

- (1) Kingdom-Animalia is characterised by heterotrophic eukaryotic organisms that are multicellular and their cells lack cell walls
- (2) Mode of nutrition in members of Kingdom Animalia is holozoic and stored food is glycogen
- (3) A few members of Kingdom-Plantae are partially heterotrophic or parasites
- (4) All of these

71. Both *Chlamydomonas* and *Chlorella* are autotrophic and both have cell wall whereas both *Paramecium* and *Amoeba* are heterotrophic and both lack cell wall, then why R.H. Whittaker place all of them under the same kingdom ?

- (1) Because all of them are single celled organisms.
- (2) Because all of them have nuclear membrane and other membrane bound cell organelles
- (3) Because all of them have cell membrane
- (4) Both (1) & (2)

72. Choose the incorrect statement :-

- (1) The vast majority of bacteria are heterotrophs
- (2) Chemosynthetic bacteria are example of Autotrophs
- (3) Nucleus is present in all nitrifying bacteria
- (4) Bacterial structure is very simple and they are very complex in behaviour

73. Choose the incorrect statement of following :-

- (1) Ribosomes are the site of protein synthesis.
- (2) Pili do not play a role in motility.
- (3) Inclusion bodies are bounded by membrane.
- (4) Cell wall prevent the bacterium from bursting and collapsing.

70. निम्न में से सत्य कथन चुनिए :-

- (1) जगत-एनिमेलिया में विषमपोषी, यूकेरियोटीक जीव रखे गए हैं जो कि बहुकोशिकीय होते हैं और उनमें कोशिका भित्ति नहीं होती है।
- (2) जगत-एनिमेलिया के सदस्यों में पोषण की विधि प्राणीसम होती है और संचित भोजन ग्लाइकोजन होता है।
- (3) जगत-प्लान्टी के कुछ सदस्य आंशिक विषमपोषी या परजीवी होते हैं।
- (4) उपरोक्त सभी

71. *क्लेमाइडोमोनास* एवं *क्लोरेला* दोनों स्वपोषी होते हैं और दोनों में कोशिका भित्ति होती है जबकि *पैरामिशियम* एवं *अमीबा* दोनों विषमपोषी होते हैं और इन दोनों में कोशिका भित्ति अनुपस्थित होती है। फिर भी आर.एच.व्हिट्टेकर ने इन सभी को एक ही जगत में क्यों रखा ?

- (1) क्योंकि ये सभी एकल कोशिकीय जीव हैं।
- (2) क्योंकि इन सभी में केन्द्रक झिल्ली और अन्य झिल्लीबद्ध कोशिकांग होते हैं।
- (3) क्योंकि इन सभी में कोशिका झिल्ली होती है।
- (4) (1) एवं (2) दोनों

72. असत्य कथन का चुनाव करें :-

- (1) अत्यधिक संख्या में जीवाणु विषमपोषी होते हैं।
- (2) रसायनसंश्लेषी जीवाणु स्वपोषी का उदाहरण हैं।
- (3) केन्द्रक सभी नाइट्रीकारी जीवाणु में पाई जाती है।
- (4) जीवाणु संरचना में सरल व व्यवहार में जटिल होते हैं।

73. निम्न में से असत्य कथन का चुनाव करें :-

- (1) राइबोसोम प्रोटीन संश्लेषण के स्थल होते हैं।
- (2) पिलाई गमन में कोई भूमिका नहीं निभाते हैं।
- (3) अन्तर्विष्ट पिण्ड झिल्ली से घिरे होते हैं।
- (4) कोशिका भित्ति जीवाणु को फटने व निपातित होने से बचाती है।

74. Consider the following four statements (A-D) and select the option which includes all the correct ones only.
- (A) Archaeobacteria differ from other bacteria in having a different cell wall structure
- (B) Bacteria as a group show the most extensive metabolic diversity
- (C) Eubacteria have chlorophyll 'c' similar to green plants.
- (D) Mycoplasma can survive without oxygen.

Option :-

- (1) Statements B, C and D
- (2) Statements A, C
- (3) Statements B, D
- (4) Statements A, B and D
75. How many organism are included in Monera in given list ?
- (A) Slimemoulds (B) Dinoflagellates
- (C) Protozoans (D) Pseudomonas
- (E) Archebacteria (F) Basidiomycetes
- (1) 2
- (2) 4
- (3) 2
- (4) 5
76. Which of the following is not related with protista ?
- (1) Eukaryote
- (2) Membrane bound organelles
- (3) Both sexual and asexual reproduction
- (4) Kingdom protista has well define boundary regarding classification
77. Which option is not correct about members of kingdom protista ?
- (1) They have cilia or flagella
- (2) They secrete poison in water
- (3) They show nitrogen fixation
- (4) They have well defined nucleus and other membrane-bounded cell organelles

74. निम्नलिखित चार कथनों (A-D) पर विचार कीजिए और केवल सभी सही कथनों वाला एक विकल्प चुनिये।
- (A) आर्की जीवाणु तथा अन्य जीवाणु की कोशिका भित्ति की संरचना एक दूसरे से भिन्न होती है।
- (B) जीवाणु में सर्वाधिक उपापचयी विविधता पाई जाती है।
- (C) सत्यजीवाणु में हरित पादपों के समान पर्णहरित 'c' पाया जाता है।
- (D) माइकोप्लाज्मा ऑक्सीजन के बिना भी जीवित रह सकते हैं।

विकल्प :-

- (1) कथन B, C और D
- (2) कथन A, C
- (3) कथन B, D
- (4) कथन A, B और D
75. दी गई सूची में से कितने जीवों को मोनेरा में शामिल किया गया है :-
- (A) स्लाइम मोल्ड (B) डायनोफ्लेजिलेट्स
- (C) प्रोटोजोअन्स (D) स्फ्यूडोमोनास
- (E) आर्किबैक्टीरिया (D) बेसिडिओमाइसीटिज
- (1) 2
- (2) 4
- (3) 2
- (4) 5
76. निम्नलिखित में से कौन प्रोटिस्टा से संबंधित नहीं है ?
- (1) यूकैरियोटिक जीव
- (2) झिल्लीबद्ध कोशिकांग
- (3) लैंगिक तथा अलैंगिक दोनों जनन
- (4) वर्गीकरण के संदर्भ में जगत प्रोटिस्टा की सीमायें निर्धारित है।
77. कौनसा विकल्प प्रोटिस्टा जगत के सदस्यों के संदर्भ में सही नहीं है ?
- (1) इनमें कशाभिका या पक्ष्माभिकाएँ होती है।
- (2) ये जल में आविष स्रावित करते हैं।
- (3) ये नाइट्रोजन स्थिरीकरण करते हैं।
- (4) इनमें सुस्पष्ट केन्द्रक तथा अन्य झिल्लीयुक्त कोशिकांग उपस्थित होते हैं।

78. Which of the following pigment is present in Euglenoid but lack in dinoflagellates and diatoms:-

- (1) Chlorophyll 'a'
- (2) Chlorophyll 'c'
- (3) Chlorophyll 'b'
- (4) Caroteens

79. Select out the incorrect match :-

- (1) Euglenoid-fresh stagnant water
- (2) Diatoms-Marine water
- (3) Dinoflagellates-Damp soil
- (4) Slime mould-tree logs and plant waste

80. Mycelium of basidiomycetes is :

- (1) Branched and septate
- (2) Unbranched and aseptate
- (3) Branched and coenocytic
- (4) Unbranched and coenocytic

81. Identify the correct match from the columns I, II and III :-

Column I		Column II		Column III	
(1)	Albugo	(a)	Ascomycetes	(i)	Obligate parasite
(2)	Agaricus	(b)	Deuteromycetes	(ii)	Decomposer of litter
(3)	Aspergillus	(c)	Phycomycetes	(iii)	Branched and septate mycelium
(4)	Alternaria	(d)	Basidiomycetes	(iv)	edible fungus

- (1) 1-c-i, 2-a-iv, 3-a-iii, 4-b-ii
- (2) 1-c-i, 2-d-iv, 3-d-iii, 4-a-ii
- (3) 1-c-i, 2-a-iii, 3-d-iv, 4-b-ii
- (4) 1-c-i, 2-d-iv, 3-a-iii, 4-b-ii

78. निम्न में से कौनसा वर्णक युग्लीनॉयड्स में तो उपस्थित होता है लेकिन डायनोफ्लेजिलेट्स व डायएटम्स में अनुपस्थित होता है-

- (1) पर्णहरित 'a'
- (2) पर्णहरित 'c'
- (3) पर्णहरित 'b'
- (4) केरोटिन्स

79. असत्य युग्म का चयन करें-

- (1) युग्लीनॉयड-स्वच्छ स्थिर जल
- (2) डायएटम्स- लवणीय जल
- (3) डायनोफ्लेजिलेट्स- आर्द्र मृदा
- (4) अवैपक कवक - लकड़ी के पुराने स्तम्भ व पादप अपशिष्ट

80. बेसिडियोमाइसीटिज का माइसिलियम होता है :-

- (1) शाखित तथा पट युक्त
- (2) अशाखित तथा पटविहिन
- (3) शाखित तथा बहुकेन्द्रिकीय
- (4) अशाखित तथा बहुकेन्द्रिकीय

81. स्तम्भ I, II व III में से सही युग्म का चुनाव करें :-

स्तम्भ I		स्तम्भ II		स्तम्भ III	
(1)	ऐब्ल्यूगो	(a)	ऐस्कोमाइसीटीज	(i)	अविकल्पी परजीवी
(2)	एगेरिकस	(b)	ड्यूटेरोमाइसीटीज	(ii)	अपशिष्ट के अपघटक
(3)	एस्पेर्जिलस	(c)	फाइकोमाइसीटीज	(iii)	शाखित व पटयुक्त कवक जाल
(4)	आल्टरनेरिया	(d)	बेसिडियोमाइसीटीज	(iv)	खाद्य कवक

- (1) 1-c-i, 2-a-iv, 3-a-iii, 4-b-ii
- (2) 1-c-i, 2-d-iv, 3-d-iii, 4-a-ii
- (3) 1-c-i, 2-a-iii, 3-d-iv, 4-b-ii
- (4) 1-c-i, 2-d-iv, 3-a-iii, 4-b-ii

- 82.** Choose the incorrect statement about phycomycetes
- (1) Members are found in aquatic habitats
 - (2) Spores are endogenously produced in sporangium
 - (3) Asexual spore are conidia
 - (4) They show all type of syngamy
- 83.** Which statement is not associated with fungi :-
- (1) Cell wall is composed of Chitin and polysaccharides
 - (2) Show symbiotic association with Algae
 - (3) They are multicellular eukaryotic autotrophic
 - (4) They causes smut, late blight & rot diseases
- 84.** Choose the incorrect statement regarding basidiomycetes :-
- (1) Asexual reproduction with the help of fragmentation.
 - (2) Rust, smut and mushroom are related with this group
 - (3) Karyogamy and meiosis take place in the basidium.
 - (4) Commonly known forms are *Agaricus*, *Aspergillus* and *Alternaria*.
- 85.** Which of the following statement is incorrect ?
- (1) Fungi prefer to grow in cold and humid places
 - (2) Karyogamy in fungi occurs after plasmogamy
 - (3) Yeast are used to make bread and beer.
 - (4) Mycelium consist of long, slender thread like structures called fungal hyphae.
- 86.** Which one of the following is mismatched :-
- (1) *Penicillium* - Source of antibiotics
 - (2) *Albugo* - Parasitic fungi on legume
 - (3) *Neurospora* - Used in biochemical and genetic work
 - (4) *Agaricus* - Edible fungus
- 82.** फाइकोमाइसिटीज के बारे में असत्य कथन का चुनाव करे-
- (1) इनके सदस्य जलीय आवासों में पाये जाते हैं
 - (2) बीजाणु का निर्माण बीजाणुधानी में अन्तःजात रूप से होता है
 - (3) अलैंगिक बीजाणु कोनिडिया होते हैं।
 - (4) ये सभी प्रकार की सिनगेमी दर्शाते हैं
- 83.** निम्न में से कौनसा कथन कवक से सम्बन्धित नहीं है :-
- (1) कोशिका भित्ति काइटिन व पौलीसैकराइड की बनी होती है
 - (2) शैवाल के साथ सहजीवी सम्बन्ध दर्शाते हैं
 - (3) ये बहुकोशिकीय, ससीमकेन्द्रकी स्वपोषी होते हैं
 - (4) ये कण्ड, पछेती अंगमारी व सड़न (Rot) रोग उत्पन्न करते हैं
- 84.** बेसिडियोमाइसिटीज के सन्दर्भ में असत्य कथन का चुनाव करे:-
- (1) अलैंगिक जनन खण्डन की मदद से होता है।
 - (2) किट्ट, कण्ड व मशरूम इस समूह से सम्बन्धित है।
 - (3) केन्द्रक संलयन व अर्द्धसूत्रण बेसिडियम में होता है।
 - (4) सामान्य ज्ञात प्रकार है *ऐगेरिकस*, *एस्परजिलस* व *आल्टरनेरिया*।
- 85.** निम्न में से कौनसा कथन गलत है ?
- (1) कवक ठण्डे और आद्र स्थानों पर उगना पसन्द करती है।
 - (2) कवक में केन्द्रक संलयन, प्लाज्मोगैमी के बाद होता है।
 - (3) यीस्ट का उपयोग ब्रेड और बीयर बनाने में किया जाता है।
 - (4) कवकजाल पतली धागे समान संरचनाओं से मिलकर बना होता है, जिसे कवक तंतु कहते हैं।
- 86.** निम्नलिखित में से कौनसा युग्म सही सुमेलित नहीं है :-
- (1) *पेनिसिलियम* - प्रतिजैविकों का स्रोत
 - (2) *एल्ब्यूगो* - दलहनों पर परजीवी कवक
 - (3) *न्यूरोस्पोरा* - जैव रासायनिक व आनुवांशिक कार्यों में उपयोगी
 - (4) *ऐगेरिकस* - खाद्य योग्य कवक

87. **Statement-1** : Lichens are parasitic association.

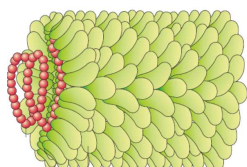
Statement-2 : In Lichens, algae prepare food for fungi and fungi provide shelter and absorb mineral nutrients and water for its partner.

- (1) Statement-1 is true, statement -2 is true; Statement -2 is not the correct explanation of statement -1
- (2) Statement-1 is true, statement -2 is true; Statement -2 is the correct explanation of statement -1
- (3) Statement-1 is true, statement -2 is false
- (4) Statement-1 is false, statement -2 is true

88. Infectious and free RNA particle having low molecular weight were discovered by :-

- (1) Diener
- (2) Beijerinck
- (3) Stanley
- (4) Pasture

89. How many statement are/is correct regarding given diagram ?



- (a) Virus has genetic material DNA.
 - (b) Virus has genetic material RNA.
 - (c) Virus has genetic material Protein.
 - (d) It causes infection in animal.
 - (e) They are living in a dicot plant.
- (1) 2
 - (2) 3
 - (3) 4
 - (4) All

87. **कथन-1** : लाइकेन परजीवी संबंध है।

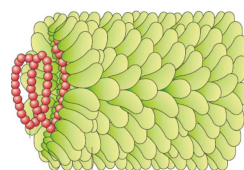
कथन-2 : लाइकेन में कवकों के लिए शैवाल भोजन बनाते हैं, तथा कवक अपने सहभागी को आश्रय देता है एवं इसके लिए खनिज पौषकों व जल का अवशोषण करता है।

- (1) कथन-1 सही है, कथन -2 सही है लेकिन कथन -2, कथन -1 का सही स्पष्टीकरण नहीं है
- (2) कथन-1 सही है, कथन -2 सही है तथा कथन -2, कथन -1 का सही स्पष्टीकरण है
- (3) कथन-1 सही है तथा कथन -2 गलत है
- (4) कथन-1 गलत है तथा कथन -2 सही है

88. कम अणुभार वाले मुक्त व संक्रमणकारी RNA के अणुओं की खोज कि गयी थी :-

- (1) डायनियर द्वारा
- (2) बेजरिनिक द्वारा
- (3) स्टेनले द्वारा
- (4) पाश्चर द्वारा

89. कितने कथन सही है दिये गये चित्र के सन्दर्भ में ?



- (a) विषाणु में अनुवांशिक पदार्थ DNA है।
 - (b) विषाणु में अनुवांशिक पदार्थ RNA है।
 - (c) विषाणु में अनुवांशिक पदार्थ प्रोटीन है।
 - (d) यह जंतुओं में संक्रमण करते हैं।
 - (e) ये एक द्विबीजपत्रीय पौधे के अन्दर जीवित रहते हैं।
- (1) 2
 - (2) 3
 - (3) 4
 - (4) सभी

90. Find out the match - that is not correct for given organisms -
- (1) *Mycoplasma* - can survive without oxygen
 - (2) *Dinoflagellates* - soap box like cell walls
 - (3) *Euglenoids* - Majority of them are fresh water organisms
 - (4) *Diatoms* - float passively on water surface
91. Which of the following is gregarious pest:
- (1) *Locusta*
 - (2) *Laccifer*
 - (3) *Aedes*
 - (4) *Limulus*
92. Which of the following structure is found in phylum Mollusca:
- (1) Radula
 - (2) Malpighian Tubules
 - (3) Nephridia
 - (4) All of the above
93. Space between mantle and visceral hump contains feather like structure called gills, function of these gill is/are:
- (1) Respiration
 - (2) Digestion
 - (3) Reproduction
 - (4) (1) and (2) both
94. Proboscis gland is present in:
- (1) *Doliolum*
 - (2) *Salpa*
 - (3) *Branchiostoma*
 - (4) *Balanoglossus*
95. Incorrect feature of *Lamprey* is:
- (1) 6-15 pair's of gill slits present
 - (2) Bony vertebral column
 - (3) Circulation is close
 - (4) Sucking and circular mouth
90. निम्नलिखित जीवों के लिए लिखे गये गलत सुमेलित को पहचानिए -
- (1) माइकोप्लाज्मा - बिना ऑक्सीजन के भी जीवित रह सकते हैं
 - (2) डाइनोफ्लेजीलेट्स - साबुनदानी के समान कोशिका भित्ति
 - (3) युग्लीनॉइड्स - अधिकांशतया स्वच्छ/ताजा (मीठे) पानी में मिलते हैं
 - (4) डायटम्स - जल की सतह पर निष्क्रिय रूप से तैरते रहते हैं
91. निम्न में से कौनसा यूथ पीड़क है:
- (1) लोकस्टा
 - (2) लेसीफर
 - (3) ऐडीस
 - (4) लीमूलस
92. निम्न में से कौनसी संरचना संघ मोलस्का में पायी जाती है:
- (1) रेड्युला
 - (2) मैलपीगीयन नलिका
 - (3) नेफ्रीडीया
 - (4) उपरोक्त सभी
93. ककुद तथा प्रावार के मध्य गुहा में पाये जाने वाले पंख समान क्लोम का कार्य बताइए:
- (1) श्वसन
 - (2) पाचन
 - (3) जनन
 - (4) (1) तथा (2) दोनों
94. प्रोबोसिस ग्रन्थि पायी जाती है:
- (1) डोलियोलम
 - (2) साल्पा
 - (3) ब्रेकिओस्टोमा
 - (4) बेलेनोग्लोसस
95. लैम्प्रे के लिए असत्य है
- (1) 6-15 जोड़ी क्लोम छिद्र उपस्थित
 - (2) अस्थिल मेरुदण्ड उपस्थित
 - (3) परिसंचरण बंद प्रकार का है
 - (4) चूसण एवं वर्तुलाकार मुख

- 96.** Repere word is related to
 (1) Reptilia
 (2) Amphibia
 (3) Aves
 (4) Mammalia
- 97.** Bird is:
 (1) Poikilothermic
 (2) Homoiothermous
 (3) Fossorial
 (4) Parasite
- 98.** Oil gland is present in:
 (1) Cockroach
 (2) Coelenterata
 (3) Aves
 (4) Sponge
- 99.** Water vascular system is the characteristic feature of:
 (1) Porifera
 (2) Echinodermata
 (3) Urochordata
 (4) Coelenterata
- 100.** Second largest phylum of animal kingdom is:
 (1) Arthropoda
 (2) Mollusca
 (3) Echinodermata
 (4) Chordata
- 101.** Type of symmetry present in Echinodermata (adult).
 (1) Asymmetry
 (2) Radial
 (3) Bilateral
 (4) None of these
- 102.** What is common between earthworm and cockroach:
 (1) Nephridia
 (2) Ventral nerve cord
 (3) Closed circulation
 (4) Malpighian tubules
- 96.** रेपरे शब्द संबंधित है
 (1) सरीसृप
 (2) उभयचर
 (3) पक्षी
 (4) स्तनधारी
- 97.** पक्षी होते हैं:
 (1) पोइकिलोथर्मिक
 (2) होमियोथर्मिक
 (3) बिलकारी
 (4) परजीवी
- 98.** तेल ग्रंथिया पायी जाती है:
 (1) कॉकरोच में
 (2) सीलेंट्रेटा में
 (3) पक्षी में
 (4) स्पंज में
- 99.** जल संवहन तंत्र किस संघ की विशेषता है:
 (1) पोरिफेरा
 (2) इकाइनोडर्मेटा
 (3) यूरोकोर्डेटा
 (4) सिलेण्ट्रेटा
- 100.** जन्तु जगत का दूसरा सबसे बड़ा संघ है:
 (1) आर्थ्रोपोडा
 (2) मोलस्का
 (3) इकाइनोडर्मेटा
 (4) कॉर्डेटा
- 101.** इकाइनोडर्मेटा (वयस्क) में पायी जाने वाली सममिती का प्रकार है:
 (1) असममिती
 (2) अरीय
 (3) द्विपाश्वरीय
 (4) उपरोक्त में से कोई नहीं
- 102.** केंचुए तथा कॉकरोच के मध्य समानता को चुनिए:
 (1) नेफ्रेडिया
 (2) अधरीय तंत्रिका रज्जु
 (3) बंद परिसंचरण
 (4) मेलपिजीयन नलिकाएं

103. Pearl is obtained from:

- (1) Asterias
- (2) *Pinctada*
- (3) Rohu
- (4) Octopus

104. Echinodermata is a group of animal which are:

- (1) Coelomate, spiny and calcareous ossicles
- (2) Acoelomate, spiny and marine calcareous ossicles
- (3) Acoelomate, horny and marine calcareous ossicles
- (4) Jointed legs, coelomate and marine calcareous ossicles

105. A flying fish is:

- (1) Rohu
- (2) Magur
- (3) Exocoetus
- (4) *Trygon*

106. A marine cartilaginous fish that can produce electric current is:

- (1) *Pristis*
- (2) *Torpedo*
- (3) *Trygon*
- (4) *Scoliodon*

107. Which of the following is true fish:

- (1) Jelly fish
- (2) Star fish
- (3) Dog fish
- (4) Devil fish

108. Scales are absent in:

- (1) Lamprey
- (2) Cartilaginous fish
- (3) Bony fish
- (4) Reptilia

103. मोती प्राप्त होते हैं:

- (1) तारा मछली
- (2) *पिंकटाडा*
- (3) रोहु
- (4) ऑक्टोपस

104. इकाइनोडर्मेटा जंतुओं का वह समुह है जो :

- (1) प्रगुहीय, शूलयुक्त तथा कैल्शियम ऑसिकल्स
- (2) अगुहीय, शूलयुक्त तथा लवणीय
- (3) अगुहीय, शूलयुक्त तथा लवणीय
- (4) संधियुक्त पाद, प्रगुहीय तथा लवणीय

105. उड़नमछली है:

- (1) रोहु
- (2) मगोर
- (3) एक्सोसीटस
- (4) *ट्राइगोन*

106. एक लवणीय उपास्थिल मछली जो विद्युतधारा उत्पन्न कर सकती है:

- (1) *प्रीस्टीस*
- (2) *टॉरपेडो*
- (3) *ट्राइगोन*
- (4) कुत्ता मछली

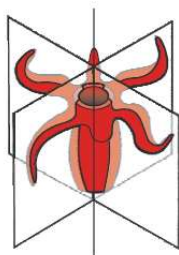
107. निम्न में से सत्य मछली है:

- (1) जेली फिश
- (2) तारामीन
- (3) कुत्ता मछली
- (4) डेवील फिश

108. शल्क अनुपस्थित होते हैं:

- (1) लेम्प्रे
- (2) उपास्थिल मछली
- (3) अस्थिल मछली
- (4) सरीसर्प

109. Following diagram represent which type of symmetry



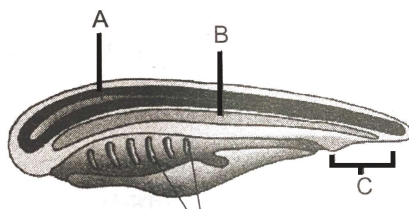
- (1) Asymmetry
- (2) Radial symmetry
- (3) Bilateral symmetry
- (4) None of these

110. The given figure represents :



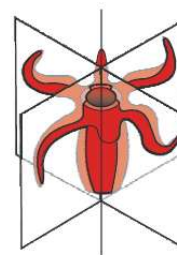
- (1) Tapeworm
- (2) Round worm
- (3) Flat worm
- (4) Both (1) and (3)

111. Choose the correct remark A,B and C in given diagram



	(B)	(A)	(C)
(1)	Notochord	Nerve cord	Post anal part
(2)	Nerve cord	Notochord	Post anal part
(3)	Notochord	Vertebral column	Gill slits
(4)	Nerve cord	Vertebral column	Gill slits

109. निम्न चित्र किस प्रकार की सममिति को दर्शाता है ?



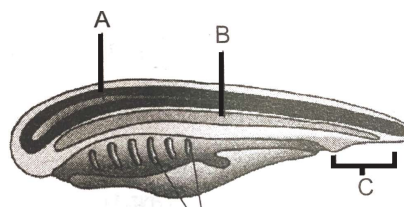
- (1) असममिति
- (2) अरीय सममिति
- (3) द्विपार्श्व सममिति
- (4) उपरोक्त में से कोई नहीं

110. दिया गया चित्र प्रदर्शित करता है ?



- (1) एक फीताकृमि को
- (2) एक गोलकृमि को
- (3) एक चपटे कृमि को
- (4) (1) तथा (3) दोनों

111. दिये गये चित्र के लिए सही नामांकित A,B व C के विकल्प चुनिए।



	(B)	(A)	(C)
(1)	नोटोकोर्ड	तंत्रिका रज्जू	पश्चगुदी पूंछ
(2)	तंत्रिका रज्जू	नोटोकोर्ड	पश्चगुदी पूंछ
(3)	नोटोकोर्ड	कशेरुक दण्ड	क्लोम दरारें
(4)	तंत्रिका रज्जू	कशेरुक दण्ड	क्लोम दरारें

112. Consider the following statements.

- (A) Mammals are exclusively marine
(B) In urochordates, notochord extends from head to tail region
(C) In cephalochordates, notochord is present only in larval tail

Which of the above statement is/are incorrect?

- (1) A alone (2) Only A and B
(3) Only B and C (4) All of these

113. Match the items in column-I with those in column-II and choose the correct option.

Column-I

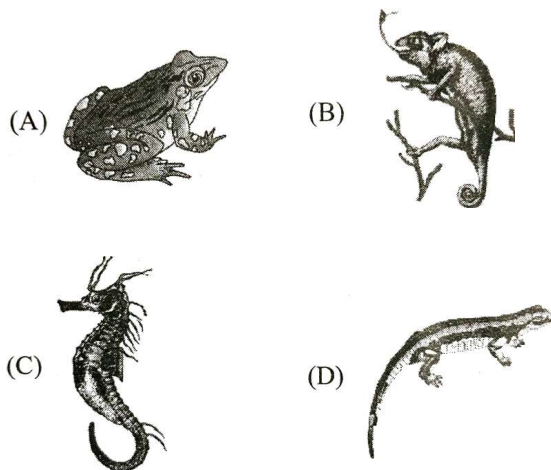
- (A) *Chelone*
(B) *Vipera*
(C) *Euspongia*
(D) *Bangarus*

Column-II

- (1) Wall lizard
(2) Viper
(3) Krait
(4) Scypha
(5) Turtle
(6) Bath sponge

- (1) A - 3, B - 2, C - 1, D - 6
(2) A - 5, B - 2, C - 4, D - 1
(3) A - 5, B - 4, C - 1, D - 6
(4) A - 5, B - 2, C - 6, D - 3

114. Which of the following are amphibians ?



- (1) A and C (2) A and D
(3) C and D (4) B and C

112. निम्नलिखित कथनों का अध्ययन कीजिये।

- (A) स्तनधारी पूर्णतया समुद्री होते हैं।
(B) यूरोकोर्डेटा में पृष्ठ रज्जू सिर से पूंछ तक फैली रहती है।
(C) सेफैलोकोर्डेटा में पृष्ठरज्जू केवल लार्वा की पूंछ में पायी जाती है।

उपरोक्त में से गलत तथ्यों को छांटिये।

- (1) सिर्फ A (2) केवल A तथा B
(3) केवल B तथा C (4) उपरोक्त सभी

113. स्तंभ-I का स्तंभ-II से मिलान करायें तथा सही विकल्प चुनिये।

स्तंभ-I

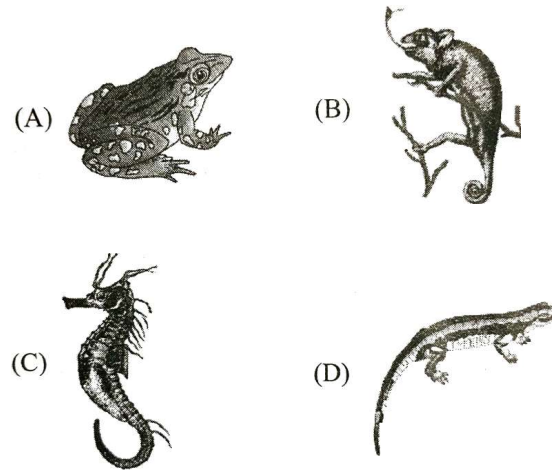
- (A) कीलोन
(B) वाइपेरा
(C) यूस्पोंजिआ
(D) बंगारूस

स्तंभ-II

- (1) छिपकली
(2) वाइपर
(3) क्रेत
(4) साइफा
(5) समुद्री कछुआ (टरटल)
(6) बाथ स्पंज

- (1) A - 3, B - 2, C - 1, D - 6
(2) A - 5, B - 2, C - 4, D - 1
(3) A - 5, B - 4, C - 1, D - 6
(4) A - 5, B - 2, C - 6, D - 3

114. निम्नलिखित में उभयचर कौन है?



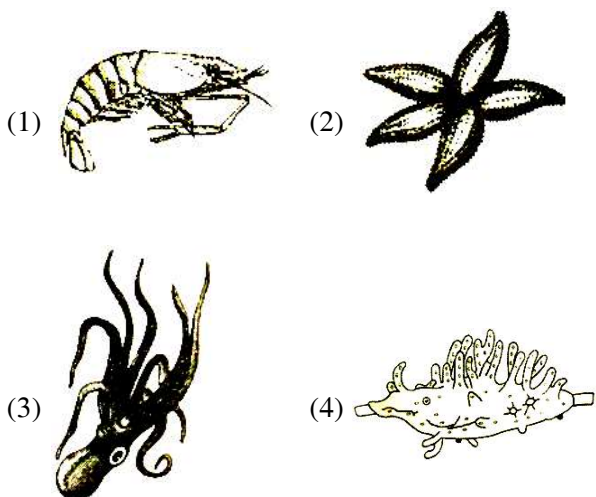
- (1) A तथा C (2) A तथा D
(3) C तथा D (4) B तथा C

115. Identify the figure with its correct name and phylum.



- (1) Balanoglossus - Urochordata
- (2) Ascidia - Urochordata
- (3) Balanoglossus - Hemichordata
- (4) Hirudinaria - Annelida

116. Which of the following animal's body is covered by calcareous shell and unsegmented with a distinct head, muscular foot, and visceral hump?



117. Flame cells present in platyhelminthes are specialized in-

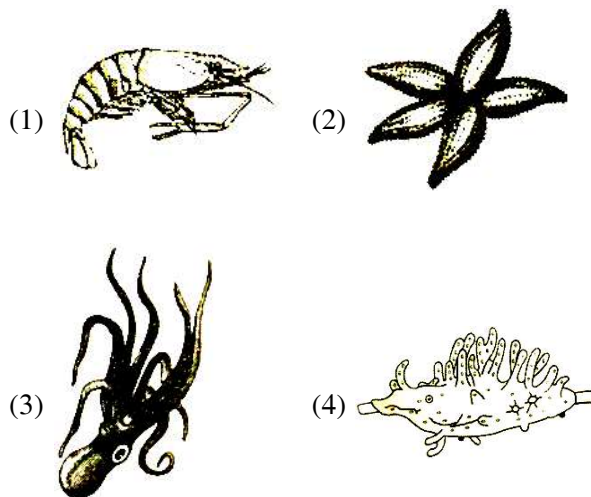
- (1) Respiration and absorption.
- (2) Osmoregulation and circulation.
- (3) Respiration and excretion.
- (4) Osmoregulation and excretion.

115. आकृति को उसके सही नाम और फाइलम में पहचानें।



- (1) बेलनोग्लोसस - यूरोकोर्डेटा
- (2) एसिडिया - यूरोकोर्डेटा
- (3) बेलनोग्लोसस - हेमीकोर्डेटा
- (4) हिरूडिनेरिया - एनेलिडा

116. निम्नलिखित में से किस जन्तु का शरीर कैल्केरियस शैल से ढका होता है और एक सिर, पेशीय पाद तथा एक अंतरंग ककुद के साथ खंडित नहीं होता है ?



117. प्लैटिहेल्मिन्थेस में मौजूद फ्लेम कोशिकाएं विशिष्ट होती हैं-

- (1) श्वसन और अवशोषण के लिए
- (2) परासरण नियमन और परिवहन के लिए
- (3) श्वसन एवं उत्सर्जन के लिए
- (4) परासरण नियमन और उत्सर्जन के लिए

118. Match the organisms given in column-I with their common name given in column-II and choose the correct option.

Column-I
(Organisms)

Column-II
(Common name)

- | | |
|--------------|------------------|
| A. Pennatula | I. Sea-fan |
| B. Gorgonia | II. Sea-pen |
| C. Adamsia | III. Sea-Anemone |
| D. Cucumaria | IV. Sea-cucumber |

- (1) A-II; B-III; C-I; D-IV
(2) A-II; B-IV; C-I; D-III
(3) A-II; B-I; C-III; D-IV
(4) A-II; B-I; C-IV; D-III

119. Which of the following possesses poison sting and belongs to class chondrichthyes?

- (1) *Labeo*
(2) *Myxine*
(3) *Clarias*
(4) *Trygon*

120. **Statement I :** Hemichordata was earlier considered as a sub-phylum under phylum Chordata.

Statement II : Balanoglossus and Saccoglossus are urochordates.

- (1) Both statements are correct
(2) Statement I is correct & II incorrect
(3) Statement I is incorrect & II correct
(4) Both statements are incorrect

121. **Statement I :** All vertebrates are chordates and all chordates are vertebrates.

Statement II : The notochord is replaced by a cartilaginous or bony vertebral column in the vertebrates.

- (1) Both statements are correct
(2) Statement I is correct & II incorrect
(3) Statement I is incorrect & II correct
(4) Both statements are incorrect

118. कॉलम-I में दिए जीवों को कॉलम-II में दिए गए उनके सामान्य नाम से मिलाएं और सही विकल्प चुनें।

कॉलम-I
(जीव)

कॉलम-II
(सामान्य नाम)

- | | |
|---------------|----------------------|
| A. पेनाटुला | I. समुंद्री पंखा |
| B. गोरगोनिया | II. समुंद्री कलम |
| C. एडमसिया | III. समुंद्री एनिमोन |
| D. कुकुमेरिया | IV. समुंद्री खीरा |

- (1) A-II; B-III; C-I; D-IV
(2) A-II; B-IV; C-I; D-III
(3) A-II; B-I; C-III; D-IV
(4) A-II; B-I; C-IV; D-III

119. निम्नलिखित में से किस किसमें जहर का डंक होता है और यह उपास्थिल क्लास से संबंधित है?

- (1) लेबियो
(2) मायक्सीन
(3) क्लारियस
(4) ट्राइगॉन

120. **कथन I :** हेमीकॉर्डेटा को पहले फाइलम कॉर्डेटा के अंतर्गत एक उप-फाइलम माना जाता था।

कथन II : बेलोनोग्लोसस और सैकोग्लोसस, यूरोकॉर्डेट्स हैं।

- (1) कथन I और कथन II दोनों सही हैं
(2) कथन I सही है लेकिन कथन II गलत है
(3) कथन I गलत है लेकिन कथन II सही है
(4) कथन I और कथन II दोनों गलत हैं

121. **कथन I :** सभी कशेरुकी रज्जुकी भी हैं, एवं सभी रज्जुकी, कशेरुकी होते हैं।

कथन II : पृष्ठरज्जु को कशेरुकियों में उपास्थिल या अस्थिल मेरुदंड द्वारा प्रतिस्थापित किया जाता है।

- (1) कथन I और कथन II दोनों सही हैं
(2) कथन I सही है लेकिन कथन II गलत है
(3) कथन I गलत है लेकिन कथन II सही है
(4) कथन I और कथन II दोनों गलत हैं

122. Statement I : Amphibia, Reptilia, Aves and Mammals included in super class tetrapoda.

Statement II : They have two pairs of limbs, adapted for walking, running, climbing, burrowing, swimming or flying.

- (1) Both statements are correct
- (2) Statement I is correct & II incorrect
- (3) Statement I is incorrect & II correct
- (4) Both statements are incorrect

123. Assertion : All members of animalia are multicellular.

Reason : All of them exhibit the same pattern of organisation of cells.

- (1) Assertion and reason both are wrong.
- (2) Assertion and reason both are true but reason in not correct explanation of assertion.
- (3) Assertion is true but reason is wrong.
- (4) Assertion and reason both are true and the reason is correct explanation of assertion.

124. Assertion : Sponges exhibit cellular level of organisation.

Reason : In sponge cells are arranged as loose cell aggregates.

- (1) Assertion and reason both are true and the reason is not correct explanation of assertion.
- (2) Assertion and reason both are true but reason in correct explanation of assertion.
- (3) Assertion is true but reason is wrong.
- (4) Assertion and reason both are wrong.

125. The members of following phylum represent tissue level of organization-

- (1) Cnidaria
- (2) Porifera
- (3) Protozoa
- (4) Both (1) and (2)

122. कथन I : उभयचर, सरीसृप, एवीज और स्तनधारी सुपर क्लास टेट्रापोडा में शामिल हैं।

कथन II : इनमें दो जोड़ी पाद हैं, जो चलने, दौड़ने, चढ़ने, बिल खोदने, तैरने या उड़ने के लिए अनुकूलित हैं।

- (1) कथन I और कथन II दोनों सही हैं
- (2) कथन I सही है लेकिन कथन II गलत है
- (3) कथन I गलत है लेकिन कथन II सही है
- (4) कथन I और कथन II दोनों गलत हैं

123. कथन : जन्तु जगत के सभी सदस्य बहुकोशिकीय हैं।

कारण : ये सभी कोशिकाओं के संगठन का एक ही पैटर्न प्रदर्शित करते हैं।

- (1) कथन और कारण दोनों असत्य कथन है।
- (2) कथन और कारण दोनों सत्य हैं लेकिन कारण कथन की सही व्याख्या नहीं है।
- (3) कथन सत्य है लेकिन कारण असत्य है।
- (4) कथन और कारण दोनों सत्य हैं और कारण कथन की सही व्याख्या है।

124. अभिकथन : स्पंज संगठन के कोशिकीय स्तर को प्रदर्शित करते हैं।

कारण : स्पंज में कोशिकाएँ ढीले कोशिका समुच्चय के रूप में व्यवस्थित होती हैं।

- (1) कथन और कारण दोनों सत्य हैं और कारण कथन की सही व्याख्या नहीं है।
- (2) कथन और कारण दोनों सत्य हैं लेकिन कारण कथन की सही व्याख्या है।
- (3) कथन सत्य है लेकिन कारण असत्य है।
- (4) कथन और कारण दोनों असत्य कथन है।

125. निम्नलिखित फाइलम के सदस्य संगठन के ऊतकीय स्तर का प्रतिनिधित्व करते हैं-

- (1) निडेरिया
- (2) पोरिफेरा
- (3) प्रोटोजोआ
- (4) दोनों (1) और (2)

126. Mark the incorrect statement for the phyla, Platyhelminthes to Echinodermata.

- (1) Groups represent organ/organ - system level of organization
- (2) All are triploblastic
- (3) All are bilaterally symmetrical (except adult echinodermata)
- (4) None of these

127. When any plane passing through the central axis of the body divides the organism into two identical halves, the organism is called-

- (1) Radially symmetrical
- (2) Bilaterally symmetrical
- (3) Asymmetrical
- (4) Metamerically segmented

128. Which of the following is a fresh water sponge?

- (1) *Sycon*
- (2) *Euspongia*
- (3) *Spongilla*
- (4) Frog

129. Comb jellies belong to-

- (1) Porifera
- (2) Cnidaria
- (3) Ctenophora
- (4) Corals

130. The phenomenon of metagenesis occur in-

- (1) *Taenia*
- (2) *Bombyx*
- (3) *Obelia*
- (4) *Musca*

126. फाइला, प्लैटिहेल्मिन्थेस से इकाइनोडर्मेटा के लिए गलत कथन को चिह्नित करें।

- (1) समूह संगठन के अंग/अंग-प्रणाली स्तर का प्रतिनिधित्व करते हैं
- (2) सभी त्रिजननस्तरीय हैं
- (3) सभी द्विपाश्वर्य रूप से सममित हैं (वयस्क इकाइनोडर्मेटा को छोड़ कर)
- (4) इनमें से कोई नहीं

127. जब शरीर के केंद्रीय अक्ष से गुजरने वाला कोई भी तल जीव को दो समान भागों में विभाजित करता है, तो जीव कहलाता है-

- (1) अरीय सममित
- (2) द्विपाश्वर्य सममिति
- (3) असममित
- (4) मेटामेरिकली खंडित

128. निम्नलिखित में से कौन सा ताजे पानी का स्पंज है?

- (1) साइकॉन
- (2) यूस्पोंजिया
- (3) स्पंजिला
- (4) मेंढक

129. कॉम्ब जेली का संबंध है-

- (1) पोरिफेरा
- (2) निडेरिया
- (3) टीनोफोरा
- (4) कोरल

130. मेटाजेनेसिस की घटना घटित होती है-

- (1) टीनिया
- (2) बॉम्बिक्स
- (3) ओबेलिया
- (4) मस्का

131. Assertion : All porifera are not living in marine water

Reason : Spongilla is a fresh water organism

- (1) Assertion and reason both are true and the reason is correct explanation of assertion.
- (2) Assertion and reason both are true but reason is not correct explanation of assertion.
- (3) Assertion is true but reason is wrong.
- (4) Assertion and reason both are wrong.

132. Assertion : Arthropoda is the largest phylum of animals which includes insects.

Reason : Over two-thirds of all named species on earth are arthropoda

- (1) Assertion and reason both are true and the reason is correct explanation of assertion.
- (2) Assertion and reason both are true and the reason is not correct explanation of assertion.
- (3) Assertion is true but reason is wrong.
- (4) Assertion and reason both are wrong.

133. Assertion : Platyhelminthes have dorso-ventrally flattened body, hence are called roundworms.

Reason : Some members like planaria possess low regeneration capacity

- (1) Assertion and reason both are true and the reason is correct explanation of assertion.
- (2) Assertion and reason both are true and the reason is not correct explanation of assertion.
- (3) Assertion is true but reason is wrong.
- (4) Assertion and reason both are wrong.

134. Acoelomate animal is.

- (1) Platyhelminthes
- (2) Annelida
- (3) Mollusca
- (4) Chordata

131. कथन : सभी स्पंज समुद्री जल में नहीं रहते हैं।

कारण : स्पोंजिला, एक स्वच्छ जलीय जन्तु है।

- (1) कथन और कारण दोनों सत्य हैं और कारण कथन की सही व्याख्या है।
- (2) कथन और कारण दोनों सत्य हैं लेकिन कारण कथन की सही व्याख्या नहीं है।
- (3) कथन सत्य है लेकिन कारण असत्य है।
- (4) कथन और कारण दोनों असत्य कथन है।

132. कारण : आर्थ्रोपोडा जानवरों का सबसे बड़ा संघ है जिसमें कीट भी शामिल हैं।

कारण : सभी नामित प्रजातियों में से दो-तिहाई से अधिक जंतु आर्थ्रोपोडा हैं।

- (1) यदि कथन और कारण दोनों सत्य हैं और कारण कथन की सही व्याख्या है
- (2) यदि कथन और कारण दोनों सत्य हैं लेकिन कारण कथन की सही व्याख्या नहीं है
- (3) यदि कथन सत्य कथन है लेकिन कारण असत्य है
- (4) यदि कथन और कारण दोनों असत्य कथन है

133. कथन : प्लैटिहेल्मिन्थिज का शरीर पृष्ठ अधरीय चपटा होता है, इसलिए इन्हें गोलकृमी कहा जाता है।

कारण : प्लेनेरिया जैसे कुछ सदस्यों में निम्न पुनरुद्भवन क्षमता होती है।

- (1) यदि कथन और कारण दोनों सत्य हैं और कारण कथन की सही व्याख्या है
- (2) यदि कथन और कारण दोनों सत्य हैं लेकिन कारण कथन की सही व्याख्या नहीं है
- (3) यदि कथन सत्य कथन है लेकिन कारण असत्य है
- (4) यदि कथन और कारण दोनों असत्य कथन है

134. अगुहीय जंतु है -

- (1) प्लैटिहेल्मिन्थीज
- (2) एनिलीडा
- (3) मोलस्का
- (4) कॉर्डेटा

135. Air bladder is present in:

- (1) Chondrichthyes
- (2) Osteichthyes
- (3) Reptilia
- (4) All fishes

135. वायु आशय पाया जाता है:

- (1) उपास्थिल मछलियों में
- (2) अस्थिल मछलियों में
- (3) सरीसृप में
- (4) सभी मछलियों में



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DATE	DAY	SYLLABUS	DATE	DAY	SYLLABUS
26-02-2024	Monday	SYLLABUS 1	08-04-2024	Monday	SYLLABUS 5 + 6
03-03-2024	Sunday	SYLLABUS 2	14-04-2024	Sunday	AIOT
09-03-2024	Saturday	SYLLABUS 3	18-04-2024	Thursday	FULL SYLLABUS
15-03-2024	Friday	SYLLABUS 4	21-04-2024	Sunday	AIOT
21-03-2024	Thursday	SYLLABUS 5	24-04-2024	Wednesday	FULL SYLLABUS
27-03-2024	Wednesday	SYLLABUS 6	28-04-2024	Sunday	FULL SYLLABUS
31-03-2024	Sunday	SYLLABUS 1 + 2	02-05-2024	Thursday	FULL SYLLABUS
04-04-2024	Thursday	SYLLABUS 3 + 4			

Proposed Syllabus 1 to 6 (Session 2023-2024)

SUBJECT ➡	PHYSICS	CHEMISTRY	BIOLOGY
SYLLABUS 1	» Physics And Measurement, » Kinematics, » Laws Of Motion, » Work, Energy, And Power » <u>Experimental Skills</u> * Vernier Calipers-Its Use To Measure The Internal And External Diameter And Depth Of A Vessel. * Screw Gauge-Its Use To Determine Thickness/Diameter Of Thin Sheet/Wire.	» Some Basic Concepts of Chemistry, » Redox reactions, » Equilibrium, » Chemical Thermodynamics	» The Living World » Biological Classification » Plant Kingdom » Structural Organisation In Animals (Animal Tissue), » Animal Kingdom » Cockroach, Frog
SYLLABUS 2	» Center Of Mass And Collision, » Circular Motion, » Rotational Motion, » Gravitation, » Properties Of Solids And Liquids, » <u>Experimental Skills</u> * Young'S Modulus Of Elasticity Of The Material Of A Metallic Wire. * Surf Ace Tension Of Water By Capillary Rise And Effect Of Detergents, * Co-Efficient Of Viscosity Of A Given Viscous Liquid By Measuring Terminal Velocity Of A Given Spherical Body	» Atomic Structure, » Classification of Elements and Periodicity in Properties, » Chemical Bonding & Molecular Structure.	» Cell : The Unit Of Life » Cell Cycle And Cell Division » Biomolecules, Enzyme » Excretory Products And Their Elimination (Excretory System) » Breathing And Exchange Of Gases (Respiratory System) » Body Fluids And Circulation (Circulatory System)
SYLLABUS 3	» Thermodynamics, » Kinetic Theory Of Gases, » Oscillations And Waves » <u>Experimental Skills</u> * Simple Pendulum-Dissipation Of Energy By Plotting A Graph Between The Square Of Amplitude And Time. * Metre Scale - The Mass Of A Given Object By The Principle Of Moments. * Speed Of Sound In Air At Room Temperature Using A Resonance Tube, * Specific Heat Capacity Of A Given (I) Solid And (II) Liquid By Method Of Mixtures.	» Some Basic Principles of Organic Chemistry, » Hydrocarbons, » Purification and Characterisation of Organic Compounds.	» Photosynthesis in Higher Plants » Plant Growth and Development » Respiration in Plants » Neural Control And Co-ordination (Nervous System) » Chemical Co-ordination And Integration (Endocrine System) » Locomotion And Movement (Muscles, Skeletal System)
SYLLABUS 4	» Electrostatics & Capacitor, » Current Electricity » <u>Experimental Skills</u> * The Resistivity of The Material of A Given Wire Using A Metre Bridge. * The Resistance Of A Given Wire Using Ohm'S Law. * Resistance And Figure Of Merit Of A Galvanometer By Half Deflection Method.	» Chemical Kinetics » Solutions » Electrochemistry » <u>Principles Related to Pratical Chemistry :-</u> The chemistry involved in the titrimetric exercises-Acids, Bases and the use of indicators, oxalic acid vs KMnO ₄ , Mohr's salt vs KMnO ₄ . Chemical principles involved in the following experiments. 1. Enthalpy of solution of CuSO ₄ . 2. Enthalpy of neutralization of strong acid and strong base. 3. Preparation of lyophilic and lyophobic sols. 4. Kinetic study of the reaction of iodide ions with hydrogen Peroxide at room temperature.	» Sexual Reproduction in Flowering Plants » Morphology Of Flowering Plants » Anatomy Of Flowering Plants » Human Reproduction » Reproductive Health,
SYLLABUS 5	» Magnetic Effects Of Current And Magnetism, » Electromagnetic Induction And Alternating Currents, » Electromagnetic Waves	» p-block elements, » d & f-block elements » Coordination Compounds » <u>Principles Related to Pratical Chemistry :-</u> 1. The chemistry involved in the preparation of Inorganic compound; Mohr's salt, potash alum. 2. Chemical principles involved in the qualitative salt analysis. 3. Cation – Pb ²⁺ Cu ⁺² Al ³⁺ Fe ³⁺ , Zn ⁺² , Ni ²⁺ Ca ²⁺ , Ba ²⁺ , Mg ²⁺ , NH ₄ ⁺ 4. Anions – CO ₃ ²⁻ , S ²⁻ , SO ₄ ²⁻ , NO ₂ ⁻ , Cl ⁻ , Br ⁻ , I ⁻ (Insoluble Salt Excluded)	» Principles Of Inheritance And Variation » Molecular Basis Of Inheritance, Microbes In Human Welfare » Biotechnology : Principles And Processes » Biotechnology And Its Applications
SYLLABUS 6	» Optics, » Dual Nature Of Matter And Radiation, » Atoms And Nuclei, » Electronic Devices » <u>Experimental Skills</u> * The Focal Length Of; (I) Convex Mirror (ii) Concave Mirror (iii) Convex Lens, Using The Parallax Method. * The Plot Of The Angle Of Deviation Vs Angle Of Incidence For A Triangular Prism. * Refractive Index Of A Glass Slab Using A Travelling Microscope. * Characteristic Curves Of A P-N Junction Diode In Forward And Reverse Bias. * Characteristic Curves Of A Zener Diode And Finding Reverse Break Down Voltage. * Identification Of Diode. Led., Resistor. A Capacitor From A Mixed Collection Of Such Items.	» Organic Compounds Containing Halogens, » Organic Compounds Containing Oxygen, » Organic Compounds Containing Nitrogen, Biomolecules » <u>Principles Related to Pratical Chemistry :-</u> 1. Detection of extra elements (Nitrogen, Sulphur, halogens) in organic compounds. 2. Detection of the hydroxyl (alcoholic and phenolic), carbonyl (aldehyde and ketones), carboxyl and amino Functional groups in organic compounds. 3. The chemistry involved in the preparation of Acetanilide, p-nitro acetanilide , aniline yellow and iodoform.	» Organisms and Populations » Ecosystem » Biodiversity and Conservation » Demography » Biology In Human Welfare : Human Health and Disease » Origin And Evolution

Read carefully the following instructions :

1. Each candidate must show on demand his/her Allen ID Card to the Invigilator.
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3. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty.
4. Use of Electronic/Manual Calculator is prohibited.
5. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Hall. All cases of unfair means will be dealt with as per Rules and Regulations of this examination.
6. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
7. The candidates will write the Correct Name and Form No. in the Test Booklet/Answer Sheet.

निम्नलिखित निर्देश ध्यान से पढ़ें :

1. पूछे जाने पर प्रत्येक परीक्षार्थी, निरीक्षक को अपना एलन पहचान पत्र दिखाए।
2. निरीक्षक की विशेष अनुमति के बिना कोई परीक्षार्थी अपना स्थान न छोड़े।
3. कार्यरत निरीक्षक को अपना उत्तर-पत्र दिए बिना कोई परीक्षार्थी परीक्षा हॉल नहीं छोड़े।
4. इलेक्ट्रॉनिक/हस्तचलित परिकलक का उपयोग वर्जित है।
5. परीक्षा हॉल में आचरण के लिए परीक्षार्थी परीक्षा के सभी नियमों एवं विनियमों द्वारा नियमित है। अनुचित साधन के सभी मामलों का फैसला परीक्षा के नियमों एवं विनियमों के अनुसार होगा।
6. किसी हालत में परीक्षा पुस्तिका और उत्तर-पत्र का कोई भाग अलग न करें।
7. परीक्षा पुस्तिका/उत्तर-पत्र में परीक्षार्थी अपना सही नाम व फॉर्म नम्बर लिखें।



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